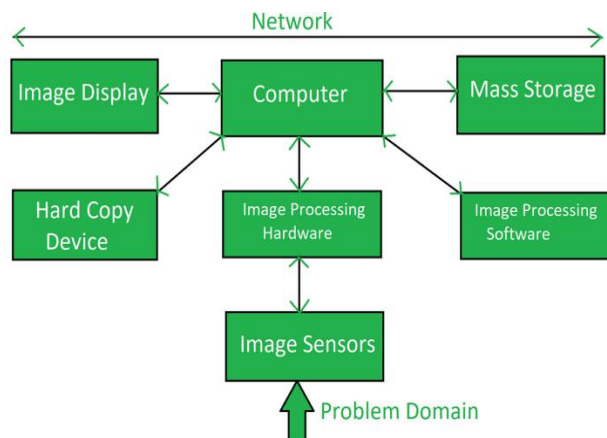


Digital Image Processing



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Cell: 018 1763 8600



Topic 2

Image Fundamentals



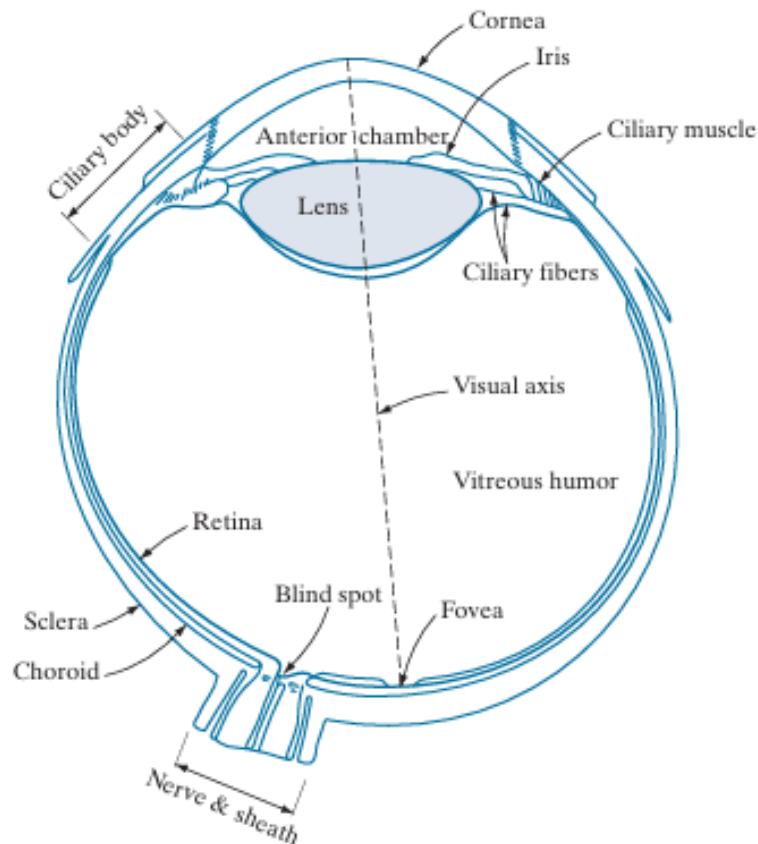
Topic Outline

- Mechanics of HVS
- Imaging Characteristics of EM spectrum
- Imaging Sensors
- Sampling and Quantization
- Representation in terms of Pixels
- Elementary Mathematical Operations



Human Visual System (HVS)

- Visual perception plays a key role in image processing
- Compare HVS and Imaging Device –resolution, adaptation to illumination change



Key components of an eye –

- Cornea –Transparent tissue that protects the eye
- Sclera –Opaque membrane of optic globe
- Choroid –Contains blood vessels. Extremely pigmented to protect backscatter inside the optic globe

Cross-section of human eye

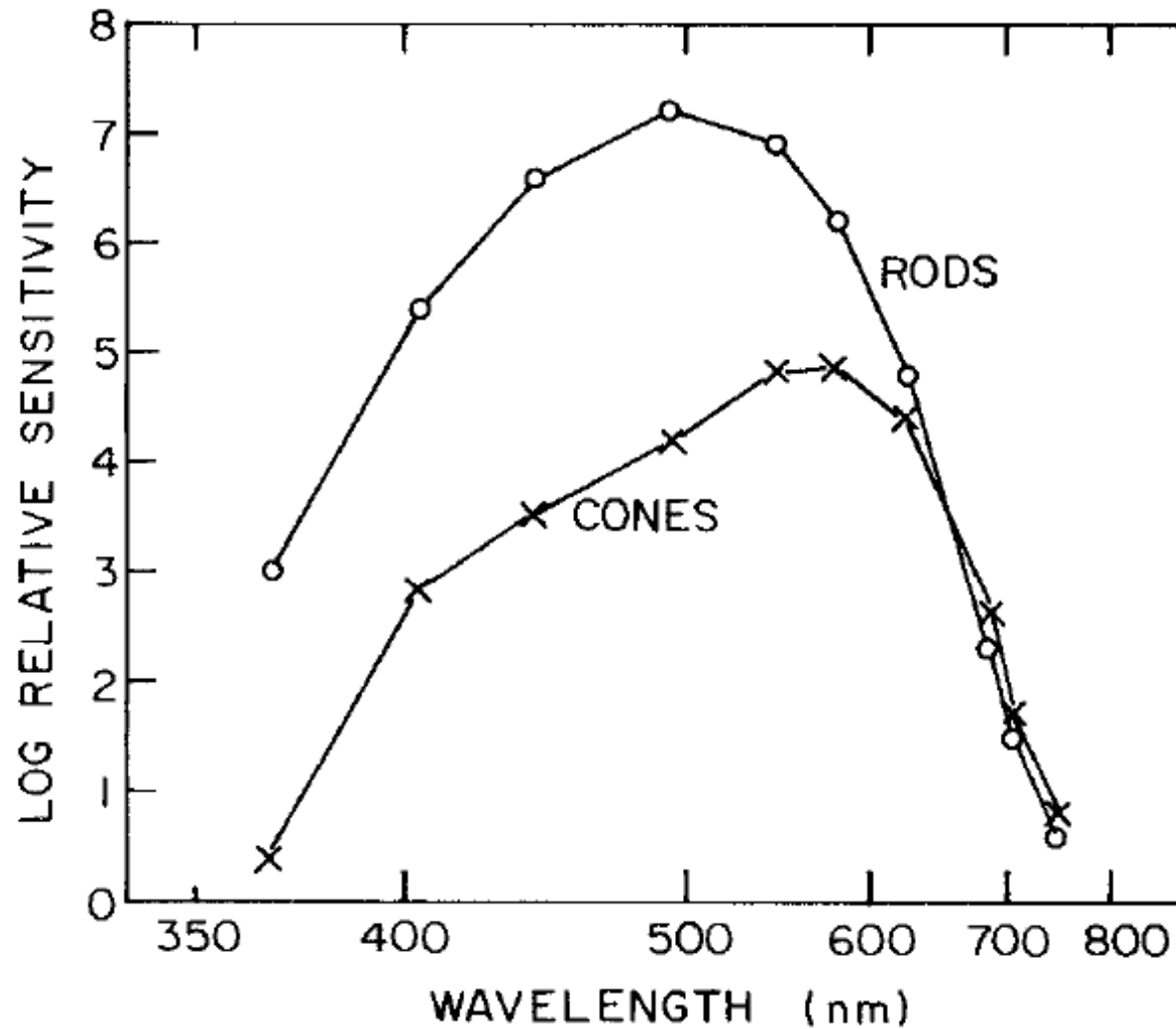


Key components of an eye –

- Ciliary body –Adjust focal length of optical lens
- Iris –control the amount of light entering the optic globe
- Lens –Concentric layers of fibrous cells. Absorbs 8% of visible spectrum with high absorption at short wavelength. Absorption in Ultraviolet or Infra red light may damage eye.
- Retina –Consists of two receptors : cones and rods.
- Cone Receptors –Low in number. Concentrated on a central position called fovea. Highly sensitive to color. Fine details response since each cone connected with a nerve.
- Rod Receptors –High in number. Distributed over the optic globe. Sensitive to low light vision with no color information. Low resolution response since several rods connected with a nerve.



Sensitivity of Rods and Cones





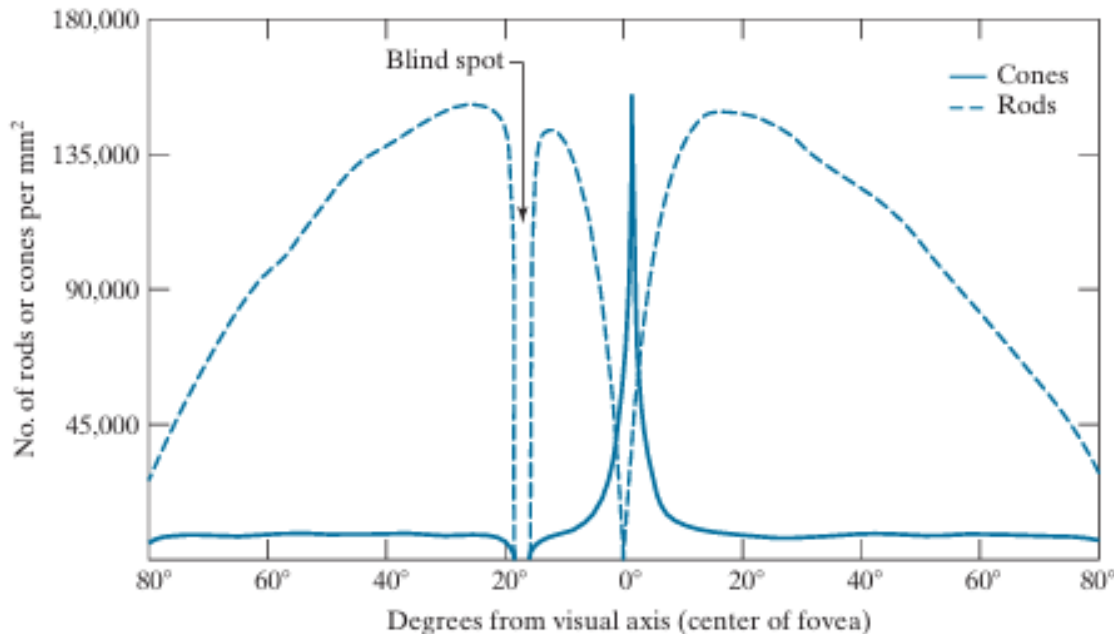
Vision Using Rods and Cones

Vision –

Cone vision: photopic / bright-light vision

Rod vision: scotopic / dim-light vision

Blind Spot – There is no receptor at the emergence of optic nerves

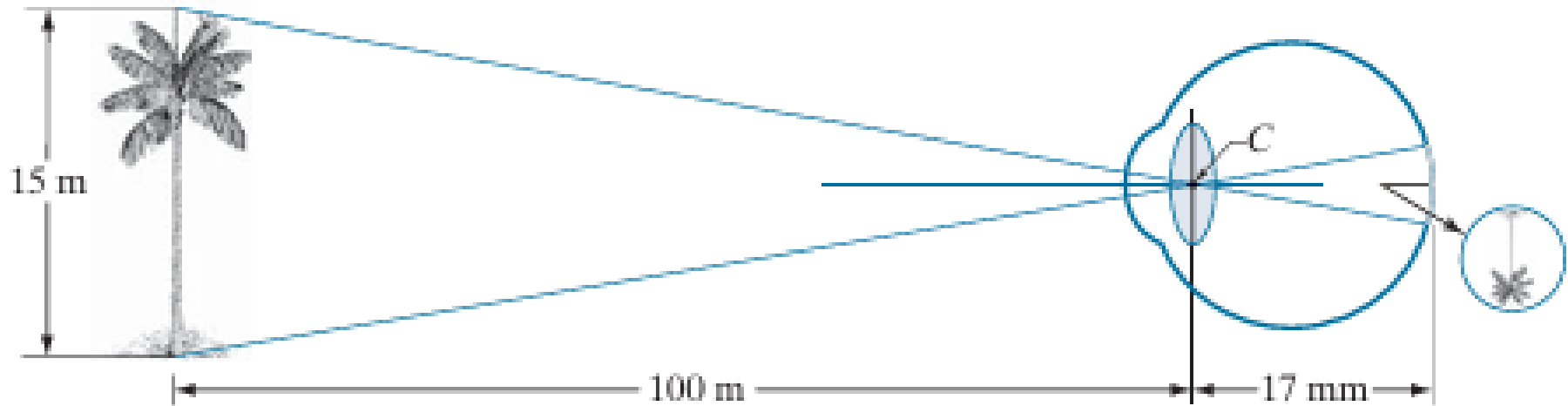


The density of cones in fovea (1.5 mm × 1.5 mm) is equivalent to resolution power of CCD sensors (5 mm × 5 mm).

HVS integrates intelligence and experience !!



Image Formation in Eye



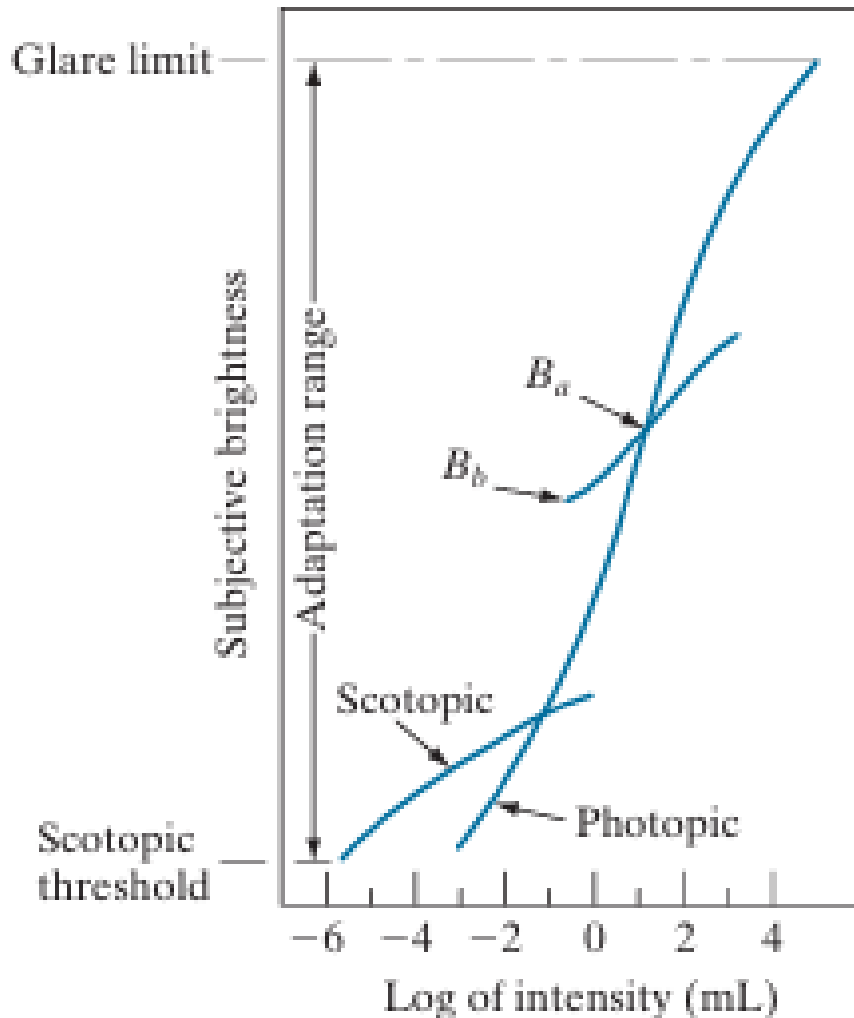
Eye lens is flexible as opposed that in camera

Focal length of eye varies in between 14 mm –17 mm
(20 mm diameter of optic globe)

Perception – (i) radiant energy to electrical impulses
(ii) decoded impulses by brain



Brightness Adaptation



- Subjective brightness is a logarithmic function of light intensity
- Although adaptation range for photopic vision is quite high, overall adaptation due to scotopic vision is quite low.
- Thus, for a given set of brightness (say B_a) condition, the range of discriminative intensity level of HVS is rather small (shown as curve of B_a - B_b).

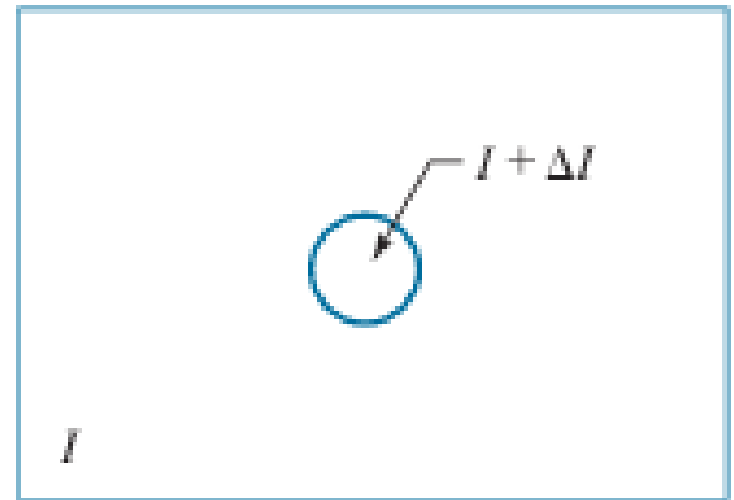


Brightness Discrimination

Weber Ratio = $\Delta I_c / I$, where ΔI_c is the increment of illumination that is 50% time discriminable from the background illumination of I .

A smaller Weber ratio means that only a small intensity change is distinguishable, i.e., good brightness discrimination.

A higher Weber ratio means that high intensity change is required to distinguish, i.e., poor brightness discrimination



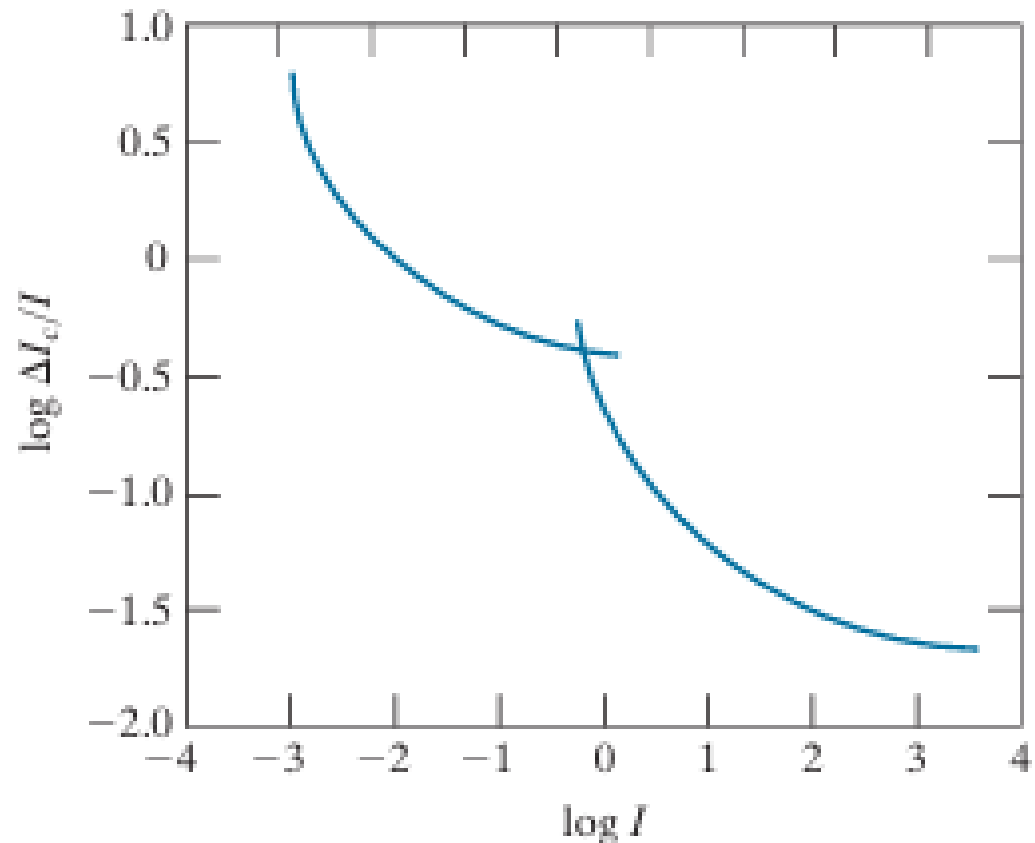
Increment of illumination
on a background



Brightness Discrimination

The power of brightness discrimination increases with the background intensity level

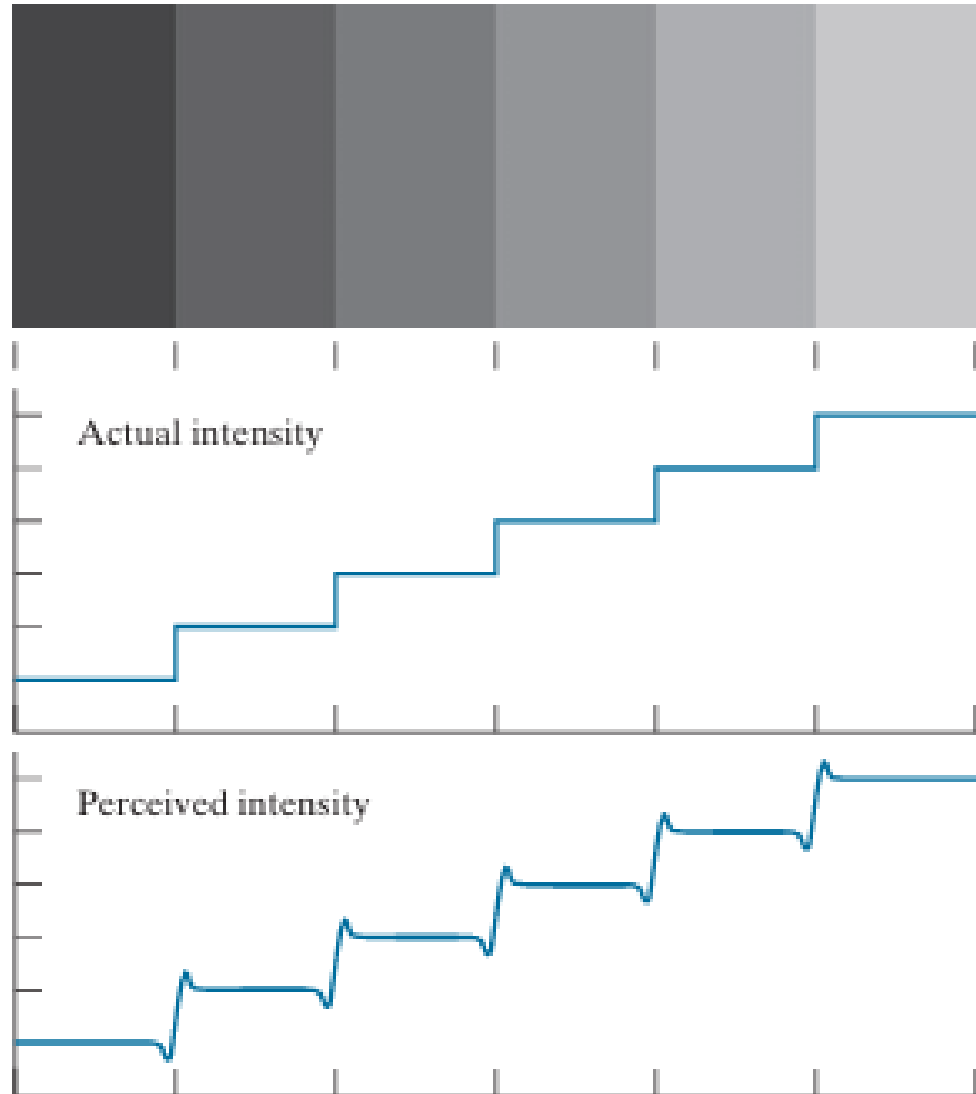
Weber ratio
as a
function of
Intensity





Brightness Discrimination (Match Bands)

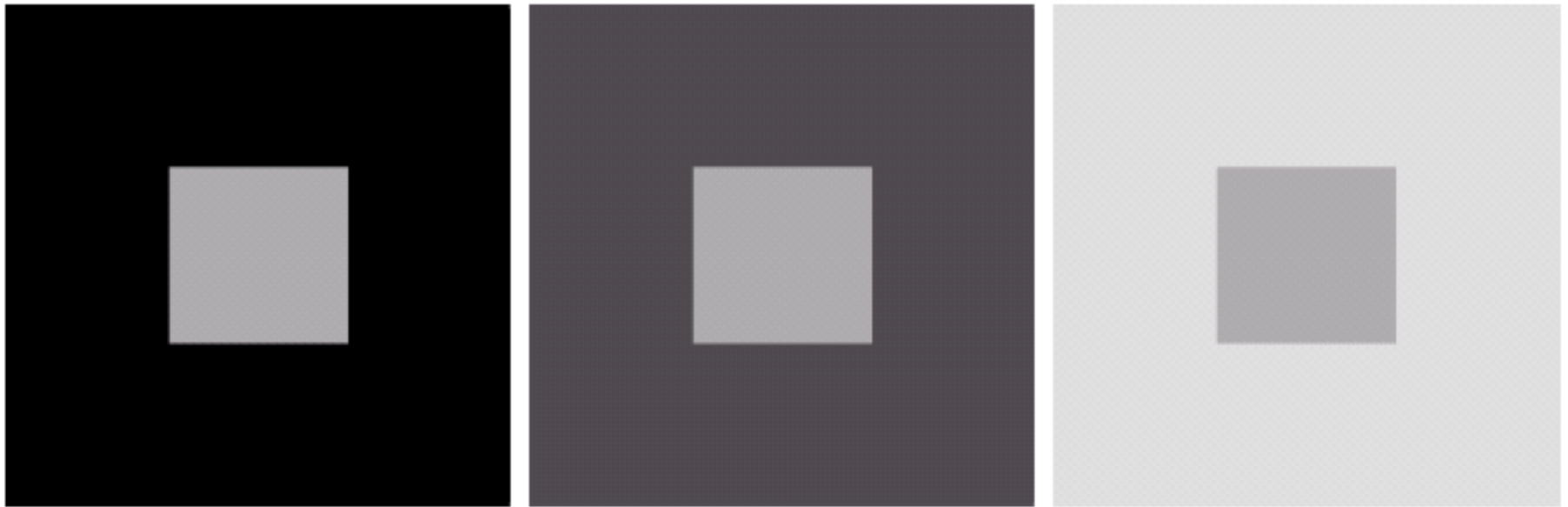
The perception of brightness tends to overshoot or undershoot around the boundary of different intensities.





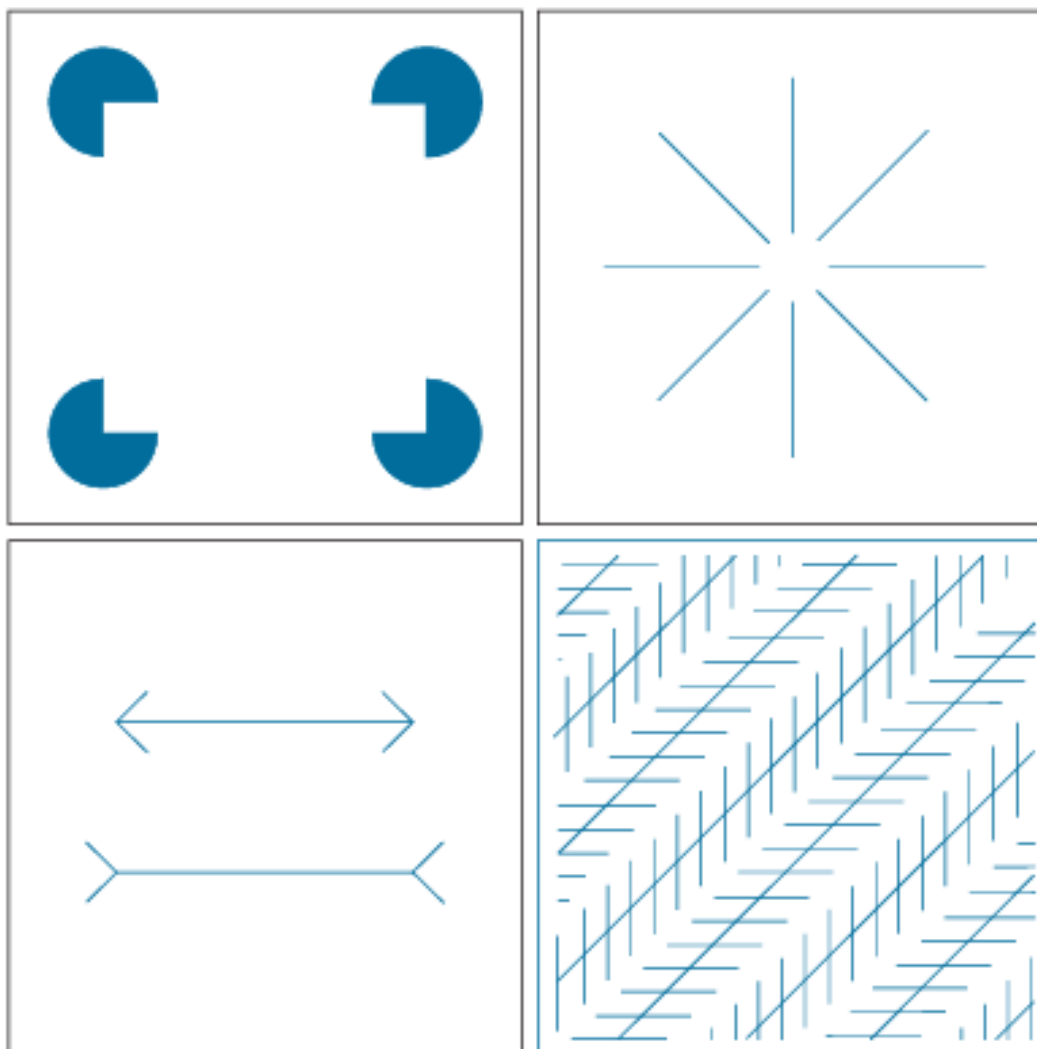
Brightness Discrimination (Simultaneous Contrast)

Same intensity level appears to be progressively darker as the background intensity increases.



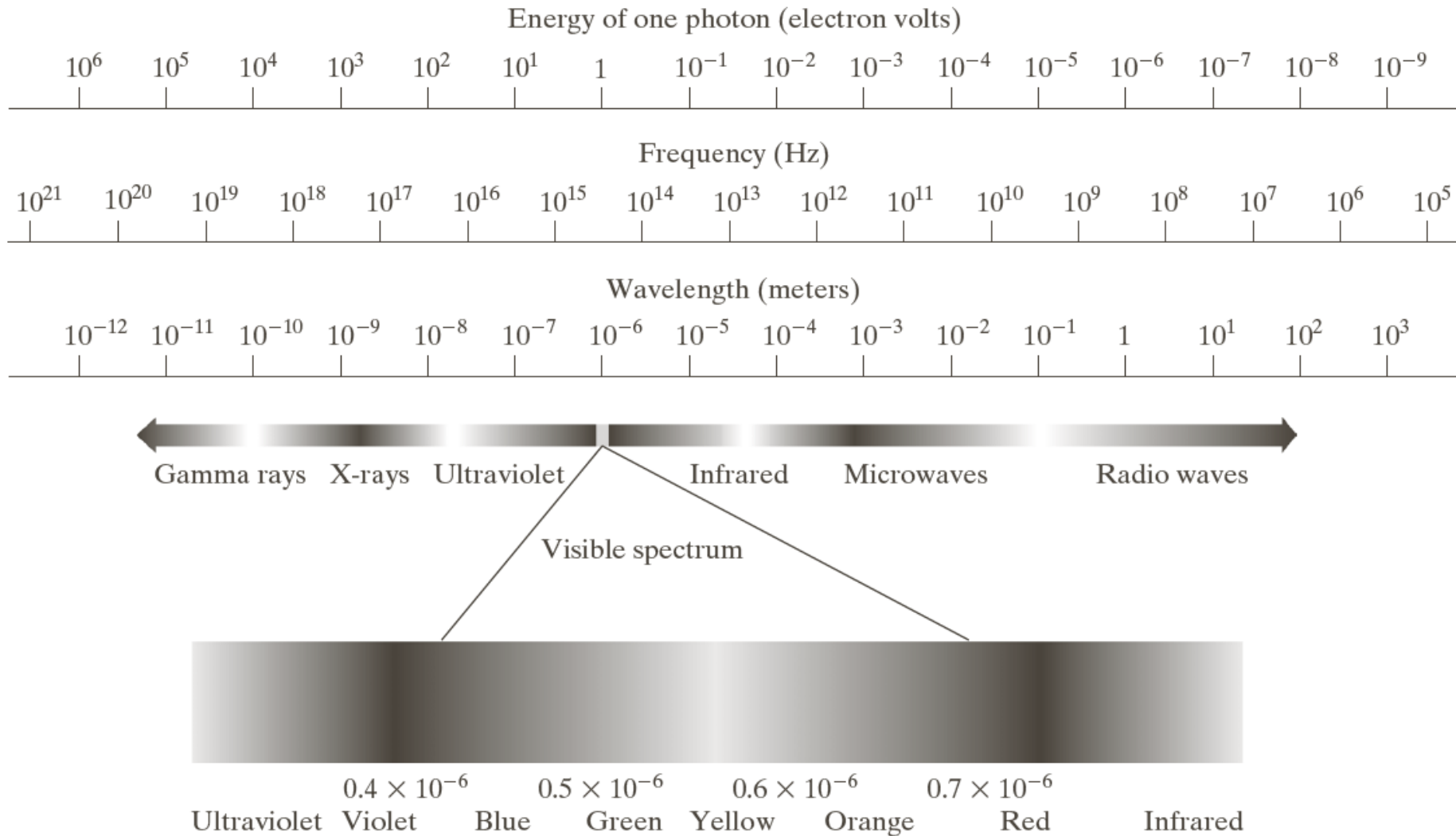


Common Optical Illusions



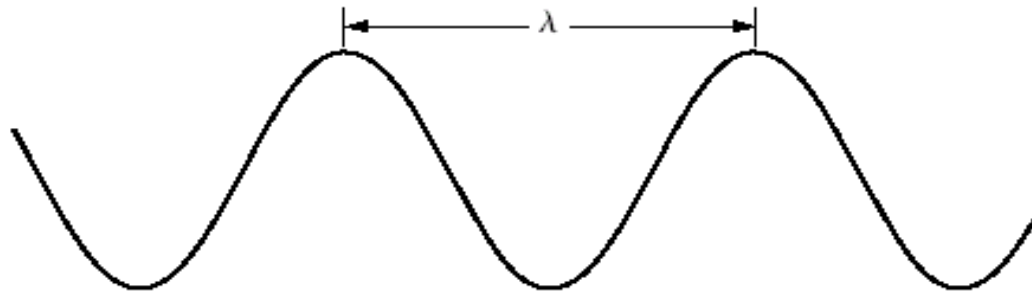


Imaging: EM Spectrum Characteristics





Imaging



- Frequency, $\nu = c/\lambda$ $c = 2.998 \times 10^8$ m/s λ = wavelength
- Energy in terms of photon $E = h\nu$ (ev), h = Planck's constant
- Gamma and Hard X-rays have high energy and hence dangerous for living organisms. Soft X-rays is used for Chest.
- Infra red is used for heat signature
- To see an object using EM spectrum, the condition $\lambda \leq D_o$ where D_o is the size of the object.
- To visualize water molecule having size 10^{-10} meter, which EM wave is required?



Imaging

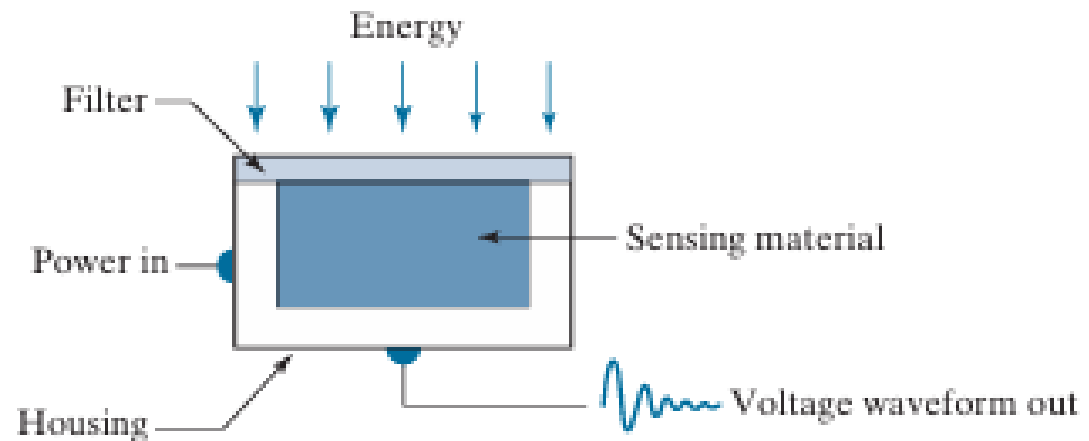
- ▶ The colors that humans perceive in an object are determined by the nature of the light reflected from the object. Green objects reflect light with wavelengths primarily in the 500 to 570 nm range while absorbing most of the energy at other wavelength
- **Monochromatic or achromatic** light –Void of color, gray level or intensity level varies from black to white. Monochromatic images are referred to as **gray-scale** images
- **Chromatic** light –Color spectrum exists. 0.43 to 0.79 μm light band. Three components, viz., radiance, luminance, and brightness
 - **Radiance** (in Watt) –Energy flows from light source
 - **Luminance** (in Lumen) –Energy perceives from light source
 - **Brightness** (subjective measure) –Energy reflects from objects which is perceived by HVS. Depends on color sensation.



Sensors

Single Image Sensor

Image sensor converts light or EM energy into electrical energy (e.g., using photo-converter for visible light)



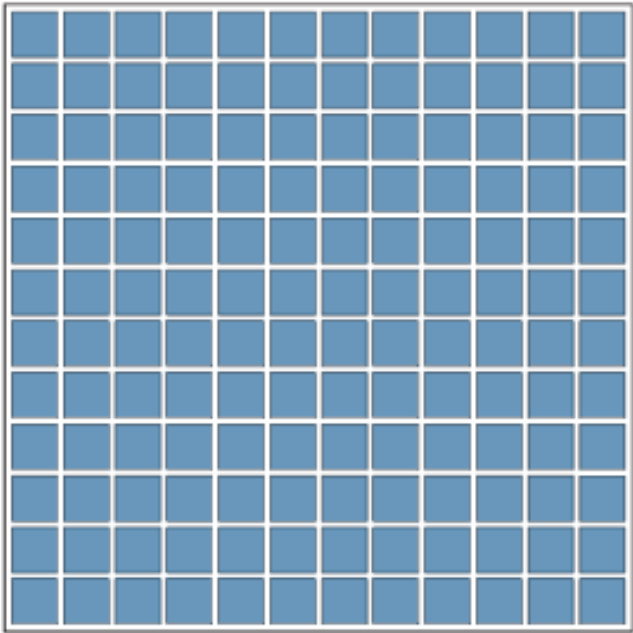


Sensors

Arrangement of Image Sensors



Line sensor

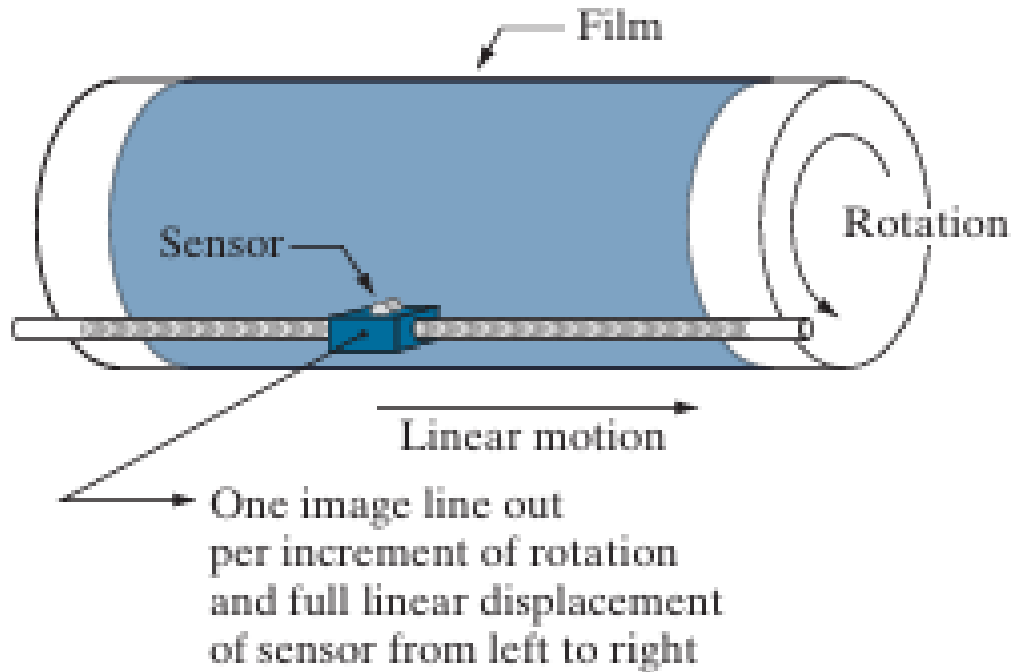


Array sensor



Image Acquisition

Using single sensor

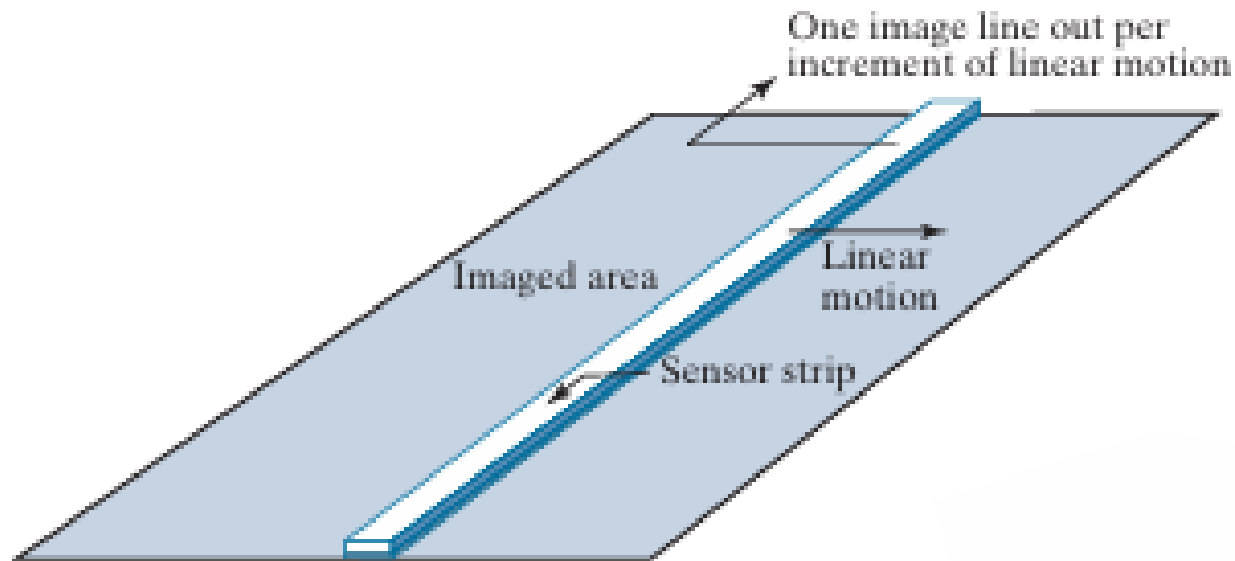


Generation of 2D image from a single sensor
Inexpensive but scanning process is slow



Image Acquisition

Using linear sensor strip



Used in air-borne applications such as generation of 2D geographical map using SAR imaging with various EM bands



Image Acquisition

Using circular sensor strip

- Used to generate 3D objects
- Require extensive processing algorithm
- Application includes CT that use
- X-ray, MRI and PET

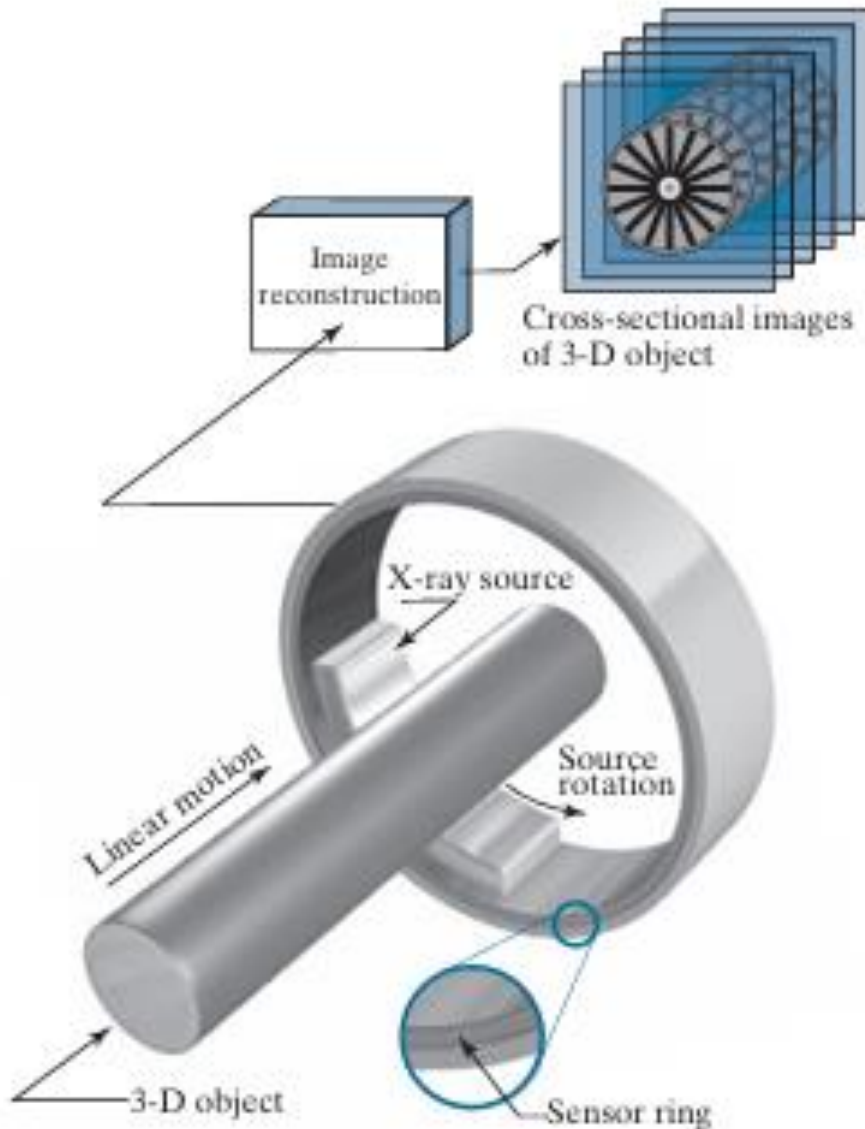




Image Acquisition

Using sensor arrays: Example includes a CCD camera

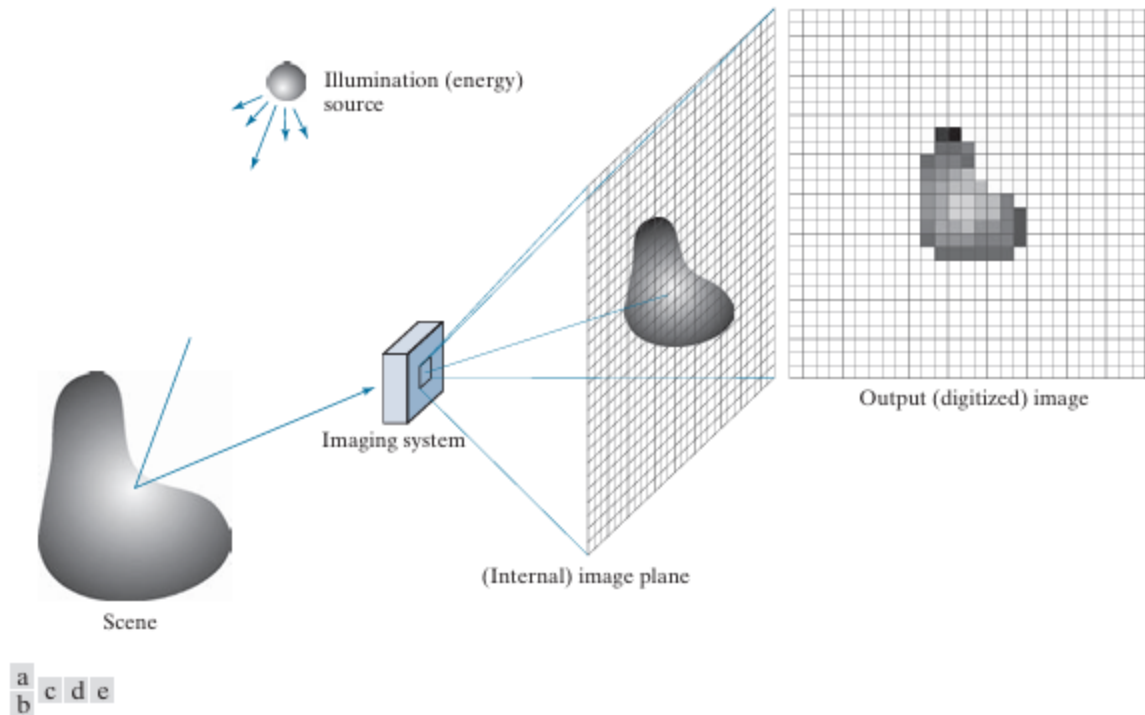


FIGURE 2.15 An example of digital image acquisition. (a) Illumination (energy) source. (b) A scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

- A typical CCD camera has 4000×4000 sensors in array
- Electrical response to each sensor $\propto$ Integral of light intensity on surface over time, and thus reducing noise
- Electrical signal is stored in digitized form as a digital image



A Simple Image Formation Model

$$f(x, y) = i(x, y) \cdot r(x, y)$$

$f(x, y)$: intensity at the point (x, y)

$i(x, y)$: illumination at the point (x, y)

(the amount of source illumination incident on the scene)

$r(x, y)$: reflectance/transmissivity at the point (x, y)

(the amount of illumination reflected/transmitted by the object)

where $0 < i(x, y) < \infty$ and $0 < r(x, y) < 1$



Image Formation

Concept of Gray Level

Image intensity $0 < f(x,y) < \infty$

where $f(x,y) = i(x,y) r(x,y)$

$$0 < i(x,y) < \infty$$

illumination

$$0 < r(x,y) < 1$$

reflectance

Intensity level: $l = f(x_0, y_0)$

where range $L_{\min} \leq l \leq L_{\max}$

$[L_{\min}, L_{\max}]$ Interval called grayscale

$[0, L-1]$ L number of discrete level

‰

0 is called black level

$L-1$ is called white level



Some Typical Ranges of Illumination

- Illumination

Lumen — A unit of light flow or luminous flux

Lumen per square meter (lm/m^2) — The metric unit of measure for illuminance of a surface

- On a clear day, the sun may produce in excess of $90,000 \text{ lm}/\text{m}^2$ of illumination on the surface of the Earth
- On a cloudy day, the sun may produce less than $10,000 \text{ lm}/\text{m}^2$ of illumination on the surface of the Earth
- On a clear evening, the moon yields about $0.1 \text{ lm}/\text{m}^2$ of illumination
- The typical illumination level in a commercial office is about $1000 \text{ lm}/\text{m}^2$



Some Typical Ranges of Reflectance

- Reflectance
 - 0.01 for black velvet
 - 0.65 for stainless steel
 - 0.80 for flat-white wall paint
 - 0.90 for silver-plated metal
 - 0.93 for snow

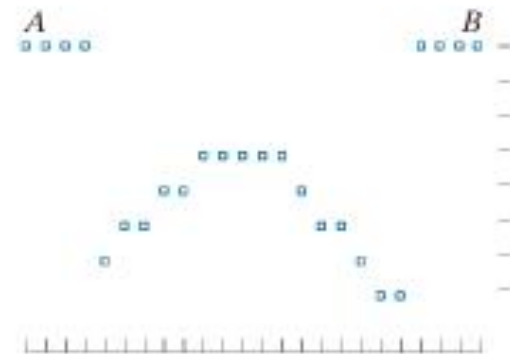
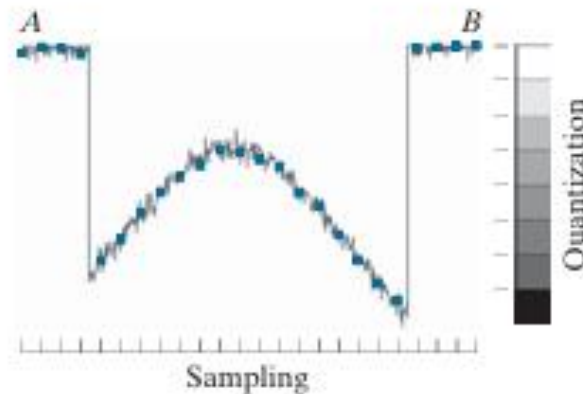
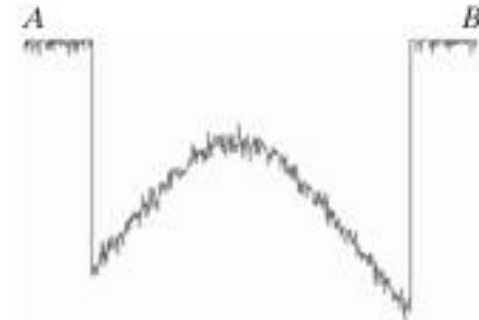
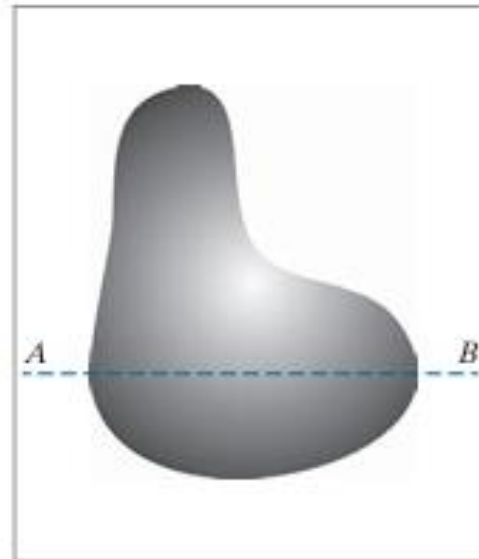


Image Sampling and Quantization

a b
c d

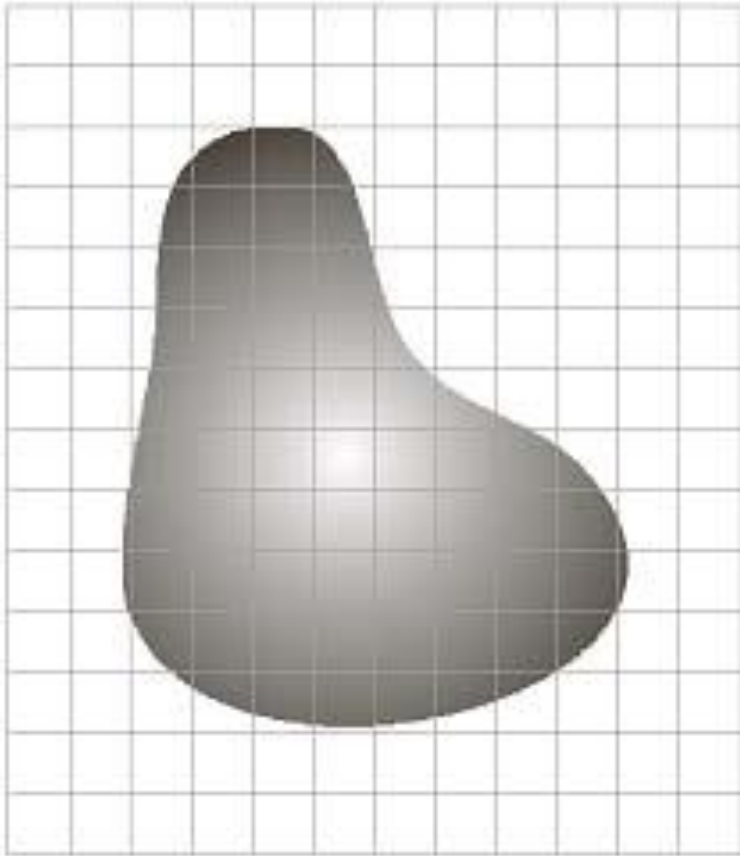
FIGURE 2.16

(a) Continuous image. (b) A scan line showing intensity variations along line AB in the continuous image. (c) Sampling and quantization. (d) Digital scan line. (The black border in (a) is included for clarity. It is not part of the image).

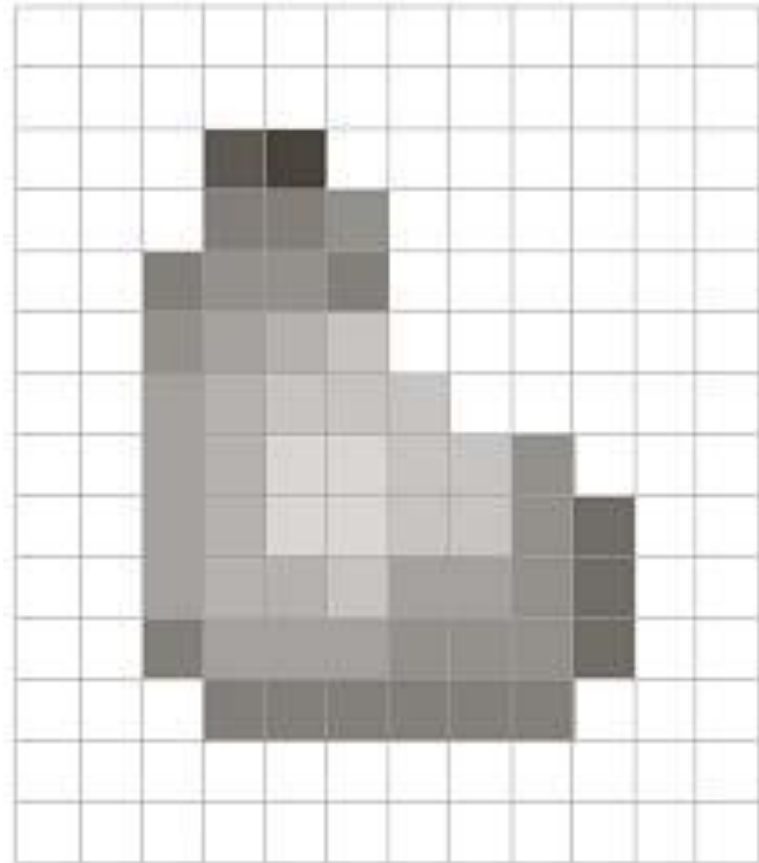




Sampling Grid



Continuous Image Grid



Discrete Image Grid



Image Representation: Matrix

a
b c

FIGURE 2.18

(a) Image plotted as a surface.
(b) Image displayed as a visual intensity array. (c) Image shown as a 2-D numerical array. (The numbers 0, .5, and 1 represent black, gray, and white, respectively.)

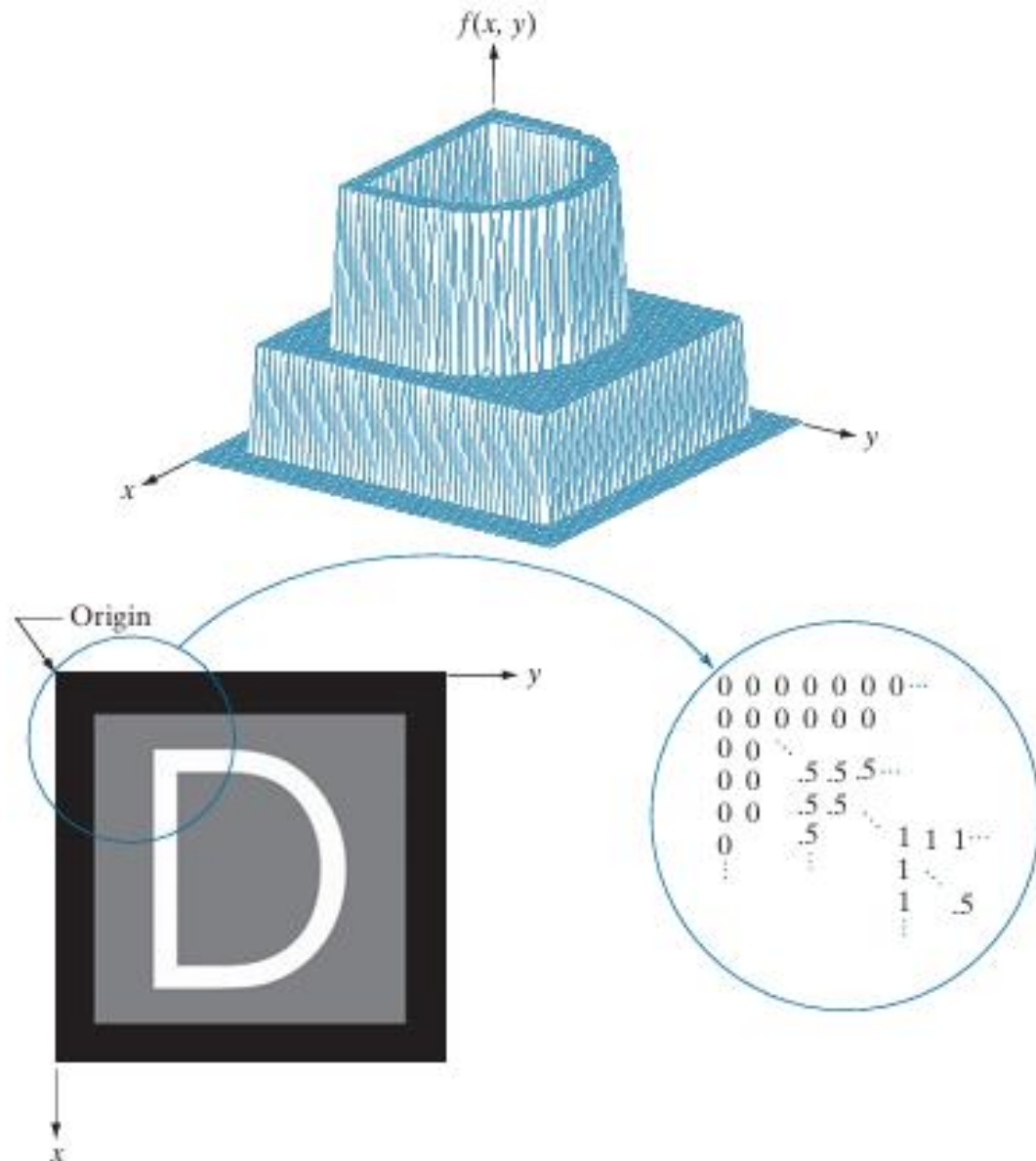


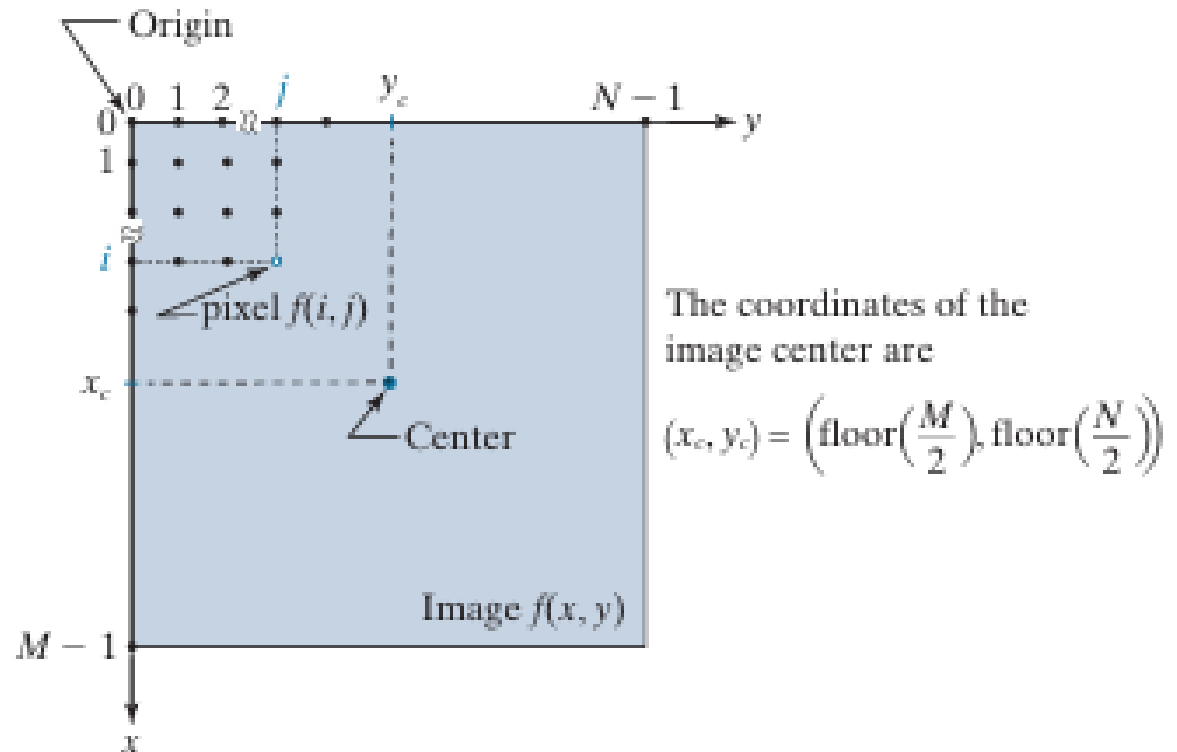


Image Coordinates

Image-element / picture-element / pixel / pel

FIGURE 2.19

Coordinate convention used to represent digital images. Because coordinate values are integers, there is a one-to-one correspondence between x and y and the rows (r) and columns (c) of a matrix.



Coordinate convention for $M \times N$ image



Representing Digital Images

- The representation of an $M \times N$ numerical array as

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \dots & f(0, N-1) \\ f(1, 0) & f(1, 1) & \dots & f(1, N-1) \\ \dots & \dots & \dots & \dots \\ f(M-1, 0) & f(M-1, 1) & \dots & f(M-1, N-1) \end{bmatrix}$$



Representing Digital Images

- The representation of an $M \times N$ numerical array as

$$A = \begin{bmatrix} a_{0,0} & a_{0,1} & \dots & a_{0,N-1} \\ a_{1,0} & a_{1,1} & \dots & a_{1,N-1} \\ \dots & \dots & \dots & \dots \\ a_{M-1,0} & a_{M-1,1} & \dots & a_{M-1,N-1} \end{bmatrix}$$



Representing Digital Images

- The representation of an $M \times N$ numerical array in MATLAB

$$f(x, y) = \begin{bmatrix} f(1,1) & f(1,2) & \dots & f(1,N) \\ f(2,1) & f(2,2) & \dots & f(2,N) \\ \dots & \dots & \dots & \dots \\ f(M,1) & f(M,2) & \dots & f(M,N) \end{bmatrix}$$



Representing Digital Images

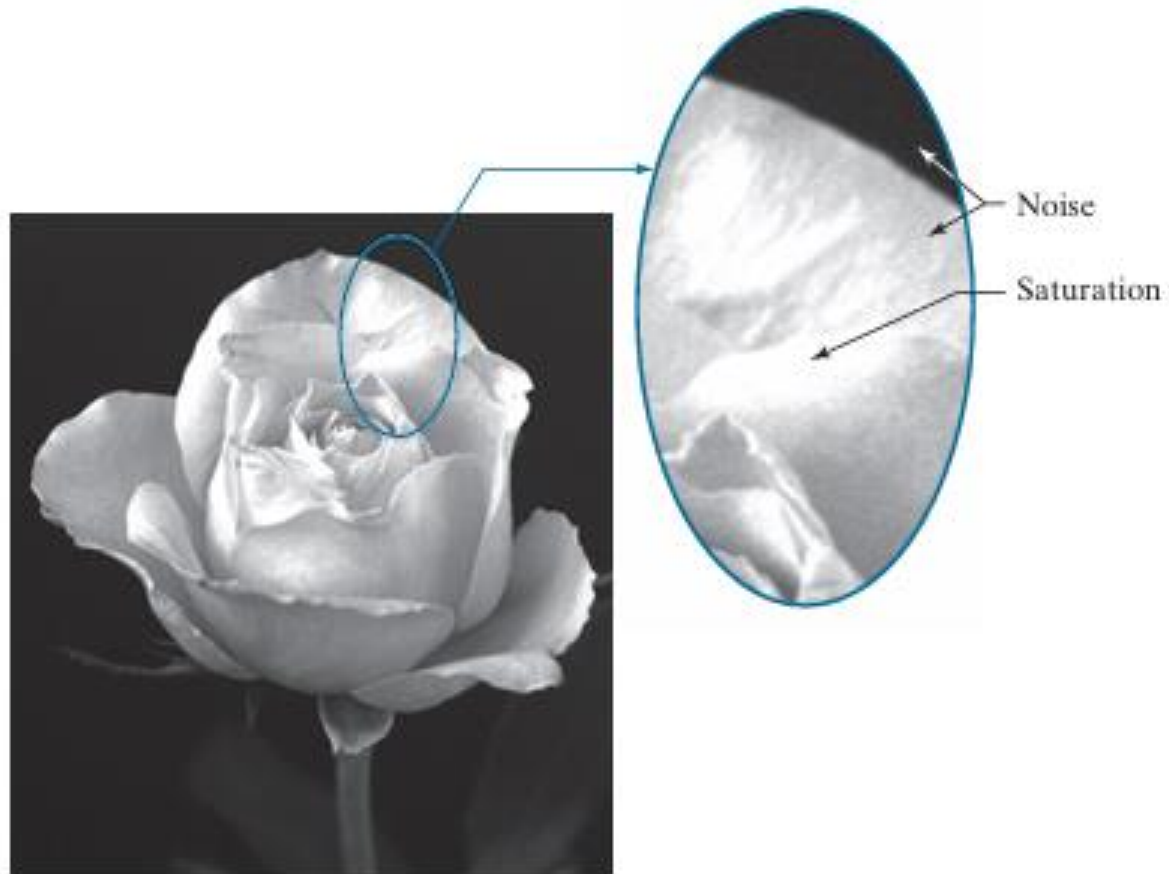
- Discrete intensity interval $[0, L-1]$, $L=2^k$
- The number b of bits required to store a $M \times N$ digitized image
 $b = MNk$
- $[0, L-1]$ Dynamic range of image
- $L = 2^k$ Number of bit depth is k , the gray level resolution



Choice of Dynamic Range

FIGURE 2.20

An image exhibiting saturation and noise. Saturation is the highest value beyond which all intensity values are clipped (note how the entire saturated area has a high, constant intensity level). Visible noise in this case appears as a grainy texture pattern. The dark background is noisier, but the noise is difficult to see.



$L_{\max}$ determines saturation level

Noise is visible as grains in the area of saturation, i.e., the brightest white area

Noise is less visible in a darker area



Storage Requirements

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

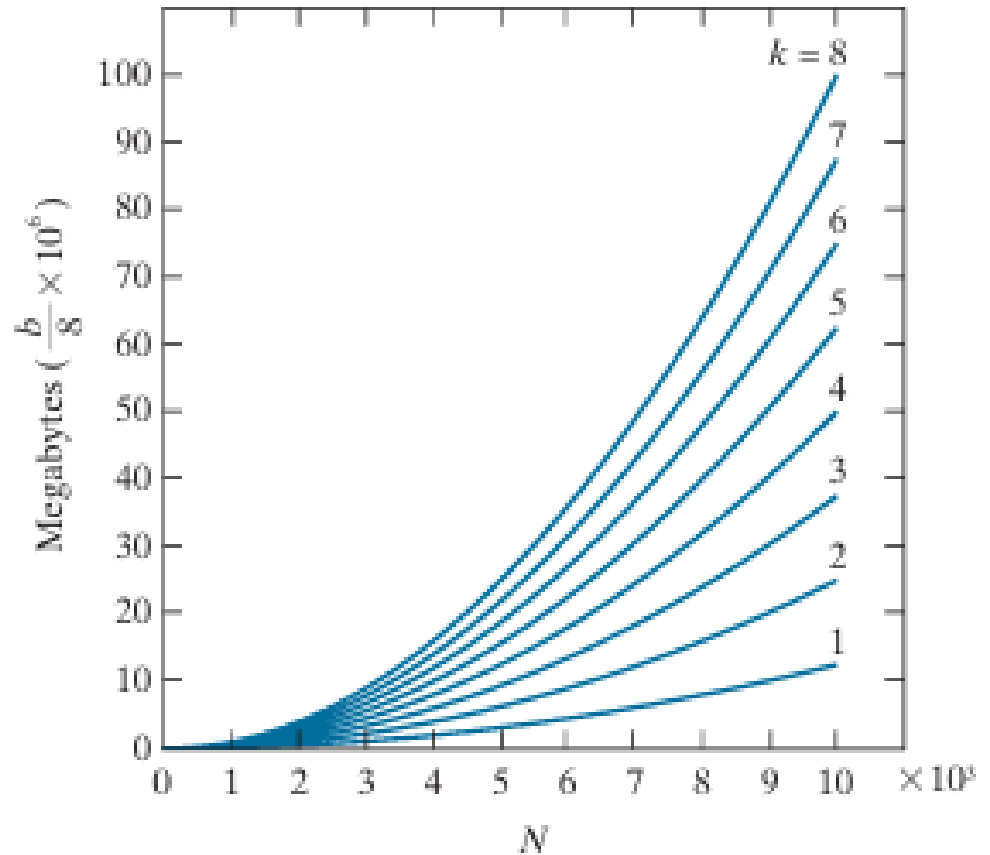
Number of bits required increases with the image size and number of quantization level



Storage Requirements

FIGURE 2.21

Number of megabytes required to store images for various values of N and k .





Spatial Resolution



Smallest number of discernible line pairs per unit distance, e.g., dot per inch (dpi) or line pairs per mm (lppm)

(a) 1250 dpi

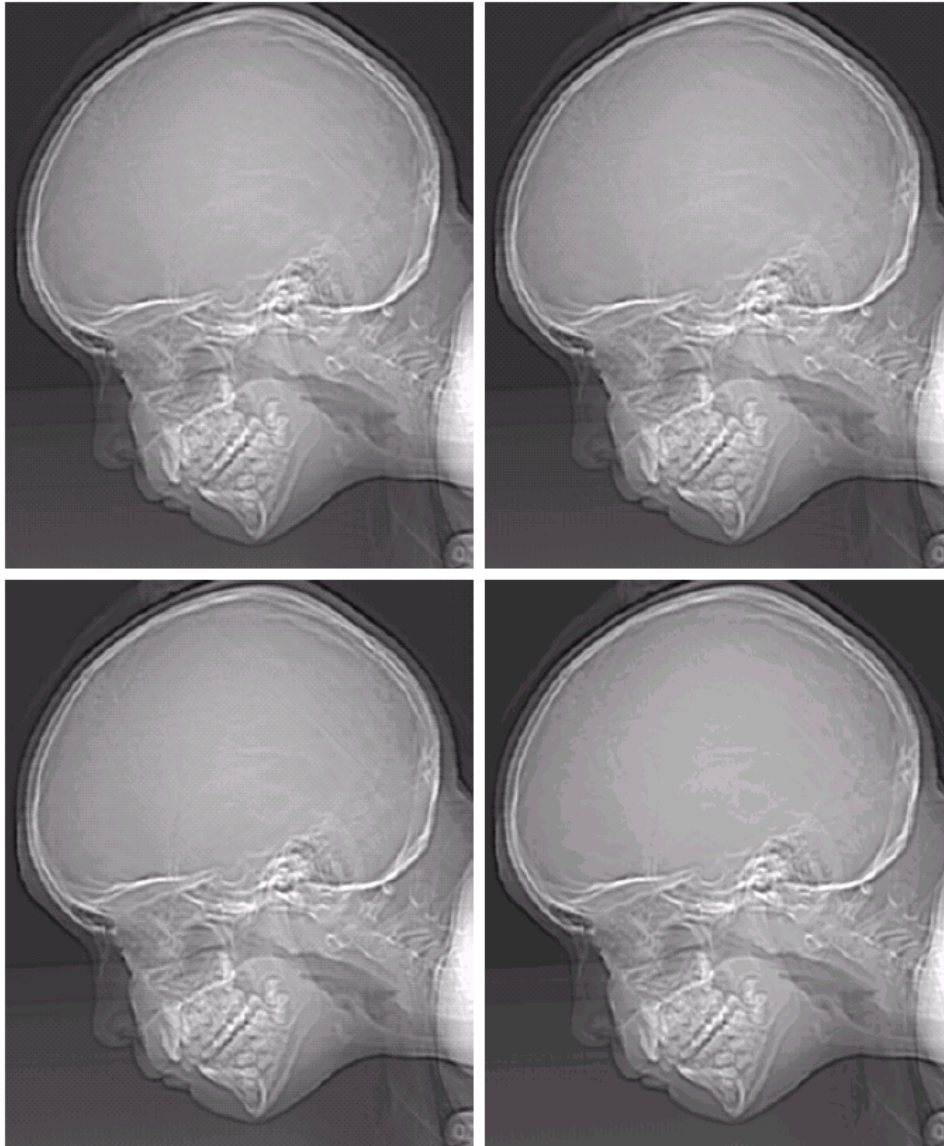
(c) 150 dpi

(b) 300 dpi

(d) 72 dpi



Gray Level Resolution

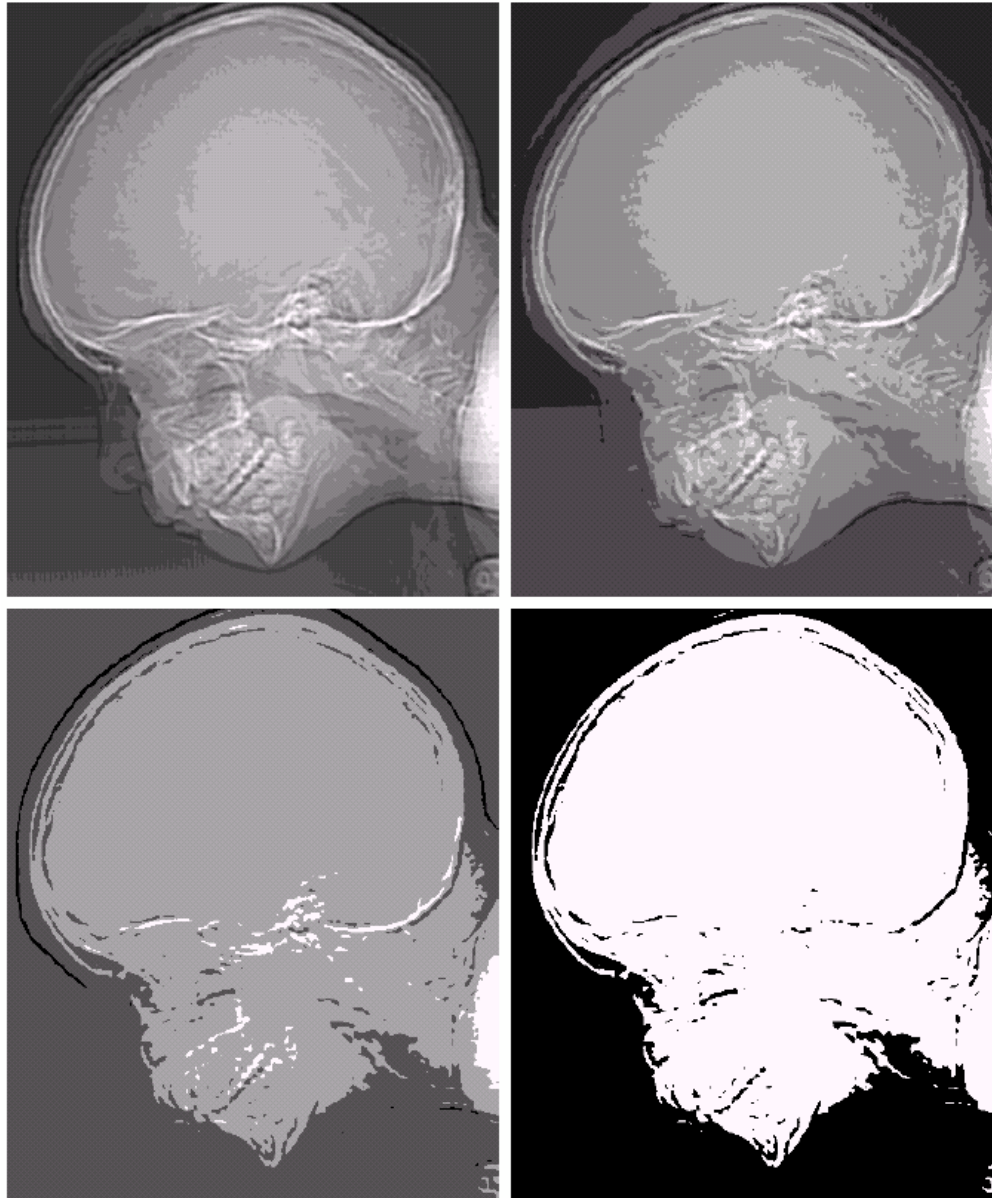


(a) 256 level (b) 128 level
(c) 64 level (d) 32 level

False contouring
starts for low level
representation



Gray Level Resolution



(a) 16 level (b) 8 level
(c) 4 level (d) 2 level

2-level representation is
a binary image



Image Details



Low detail



Medium detail

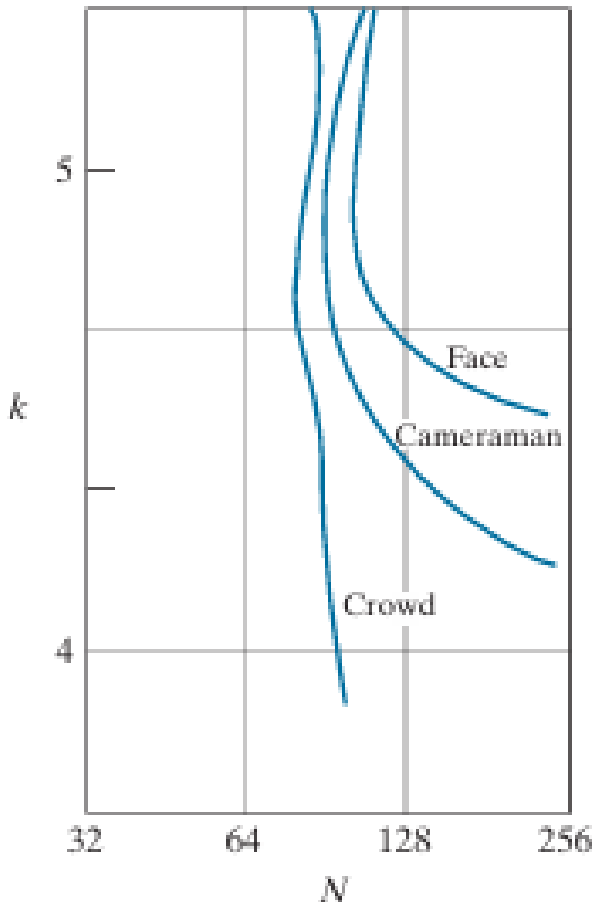


High detail



Image Quality

Isopreference curve



- Isopreference curve means equal subjective quality
- Subjective quality does not significantly depend on the bit depth of image having large details
- Subjective quality significantly depends on the bit depth of image having low details, but increased sample size may improve quality slightly



Operation on Pixels

$H(af + bg) = aH(f) + bH(g)$. Linear Operation

4-neighbors of $p \Rightarrow N_4(p)$

$(x + 1, y), (x - 1, y), (x, y + 1), (x, y - 1)$

Diagonal neighbors of $p \Rightarrow N_D(p)$

$(x + 1, y + 1), (x + 1, y - 1), (x - 1, y + 1), (x - 1, y - 1)$

8-neighbors of $p \Rightarrow N_8(p) \in N_4(p) \cup N_D(p)$

Relations between pixels are based on:

Neighborhood

Adjacency

Connectivity

Paths

Regions and boundaries



Operation (Distances)

Euclidian Distance $\Rightarrow D_e(p, q) = [(x - s)^2 + (y - t)^2]^{\frac{1}{2}}.$

Distance from center of a circle

City-block Distance $\Rightarrow D_4(p, q) = |x - s| + |y - t|.$

Distance from center of a diamond

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

Chessboard Distance $\Rightarrow D_8(p, q) = \max(|x - s|, |y - t|).$

Distance from center of a square

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2



Image Interpolation

- **Interpolation** — Process of using known data to estimate unknown values
e.g., zooming, shrinking, rotating, and geometric correction
- **Interpolation** (sometimes called *resampling*) — an imaging method to increase (or decrease) the number of pixels in a digital image.

Some digital cameras use interpolation to produce a larger image than the sensor captured or to create digital zoom



Image Interpolation:

Nearest Neighbor Interpolation

$$\begin{aligned} f_1(x_2, y_2) &= \\ f(\text{round}(x_2), \text{round}(y_2)) &= \\ = f(x_1, y_1) \end{aligned}$$

$$f(x_1, y_1)$$

$$\begin{aligned} f_1(x_3, y_3) &= \\ f(\text{round}(x_3), \text{round}(y_3)) &= \\ = f(x_1, y_1) \end{aligned}$$

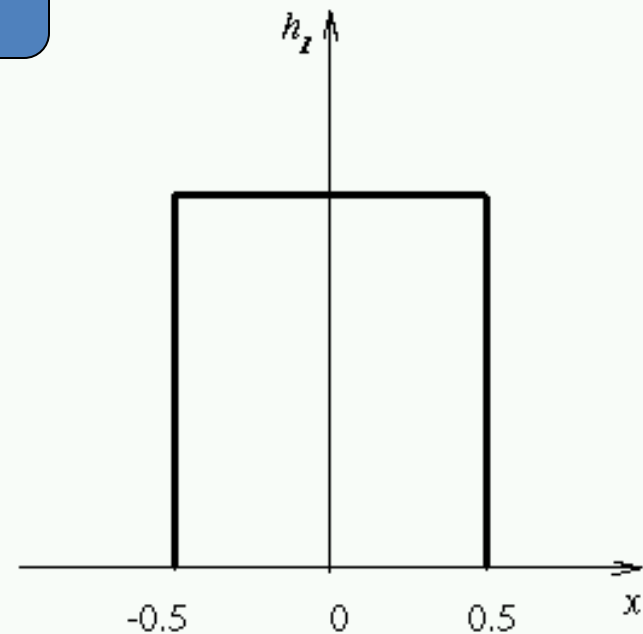
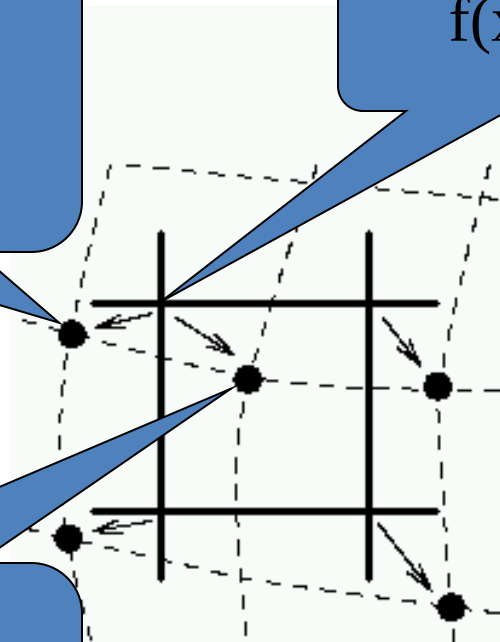




Image Interpolation:

Bicubic Interpolation

- The intensity value assigned to point (x,y) is obtained by the following equation

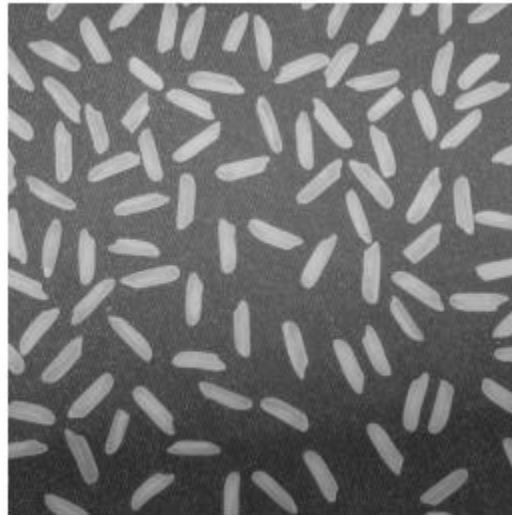
$$f_3(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

- The sixteen coefficients are determined by using the sixteen nearest neighbors.



Examples: Interpolation

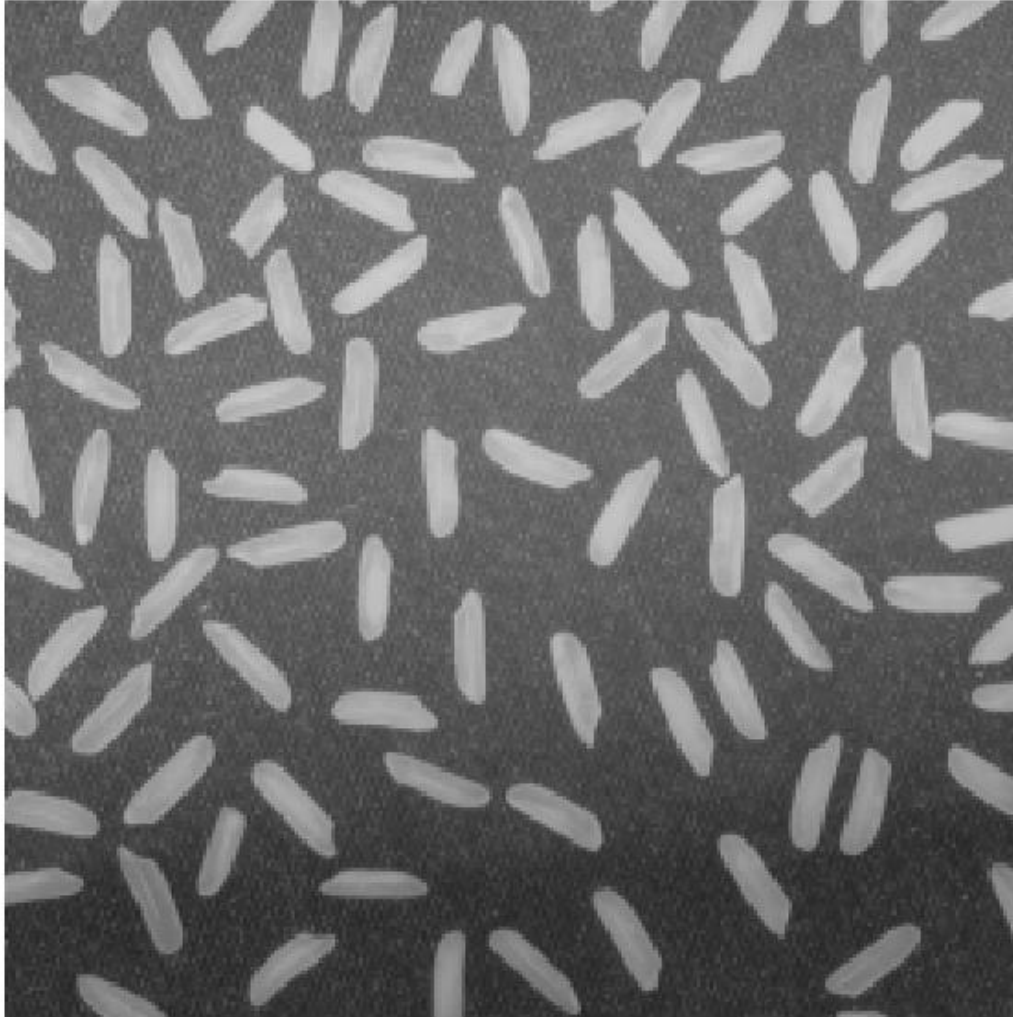
original image





Examples: Interpolation

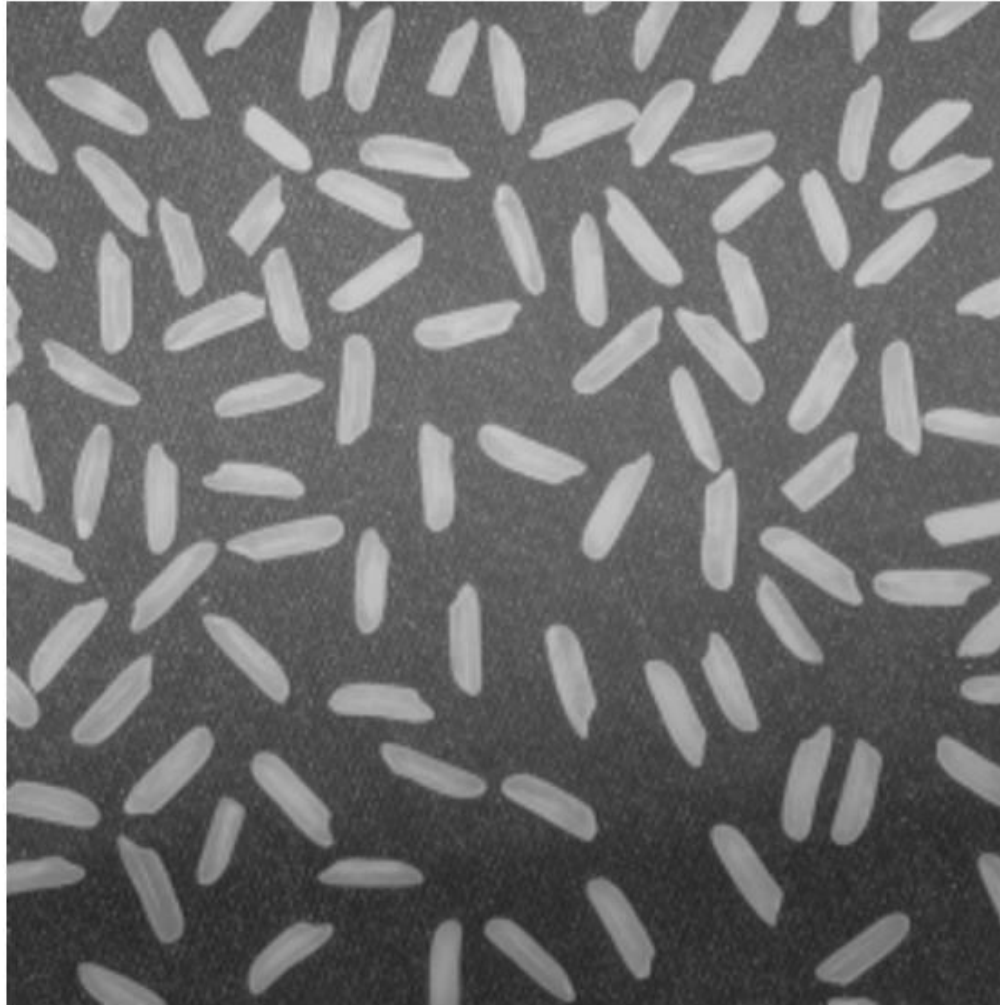
nearest





Examples: Interpolation

bicubic





Basic Relationships Between Pixels

- Neighborhood
- Adjacency
- Connectivity
- Paths
- Regions and boundaries



Basic Relationships Between Pixels

- **Neighbors** of a pixel p at coordinates (x,y)
 - **4-neighbors of p** , denoted by $N_4(p)$:
 $(x-1, y)$, $(x+1, y)$, $(x,y-1)$, and $(x, y+1)$.
 - **4 diagonal neighbors of p** , denoted by $N_D(p)$:
 $(x-1, y-1)$, $(x+1, y+1)$, $(x+1,y-1)$, and $(x-1, y+1)$.
 - **8 neighbors of p** , denoted $N_8(p)$
$$N_8(p) = N_4(p) \cup N_D(p)$$



Basic Relationships Between Pixels

- **Adjacency**

Let V be the set of intensity values

- **4-adjacency:** Two pixels p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.
- **8-adjacency:** Two pixels p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.



Basic Relationships Between Pixels

- **Adjacency**

Let V be the set of intensity values

➤ **m-adjacency:** Two pixels p and q with values from V are m-adjacent if

(i) q is in the set $N_4(p)$, or

(ii) q is in the set $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V .



Basic Relationships Between Pixels

- **Path**

- A (digital) path (or curve) from pixel p with coordinates (x_0, y_0) to pixel q with coordinates (x_n, y_n) is a sequence of distinct pixels with coordinates

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

Where (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \leq i \leq n$.

- Here n is the *length* of the path.
- If $(x_0, y_0) = (x_n, y_n)$, the path is ***closed*** path.
- We can define 4-, 8-, and m-paths based on the type of adjacency used.



Basic Relationships Between Pixels

- **Connected in S**

Let S represent a subset of pixels in an image. Two pixels p with coordinates (x_0, y_0) and q with coordinates (x_n, y_n) are said to be **connected in S** if there exists a path

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$$

Where $\forall i, 0 \leq i \leq n, (x_i, y_i) \in S$



Basic Relationships Between Pixels

Let S represent a subset of pixels in an image

- For every pixel p in S , the set of pixels in S that are connected to p is called a ***connected component*** of S .
- If S has only one connected component, then S is called ***Connected Set***.
- We call R a **region** of the image if R is a connected set
- Two regions, R_i and R_j are said to be ***adjacent*** if their union forms a connected set.
- Regions that are not to be adjacent are said to be ***disjoint***.



Basic Relationships Between Pixels

- **Boundary (or border)**
 - The **boundary** of the region R is the set of pixels in the region that have one or more neighbors that are not in R .
 - If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns of the image.
- **Foreground and background**
 - An image contains K disjoint regions, R_k , $k = 1, 2, \dots, K$. Let R_u denote the union of all the K regions, and let $(R_u)^c$ denote its complement.
All the points in R_u is called **foreground**;
All the points in $(R_u)^c$ is called **background**.



Distance Measures

- Given pixels p , q and z with coordinates (x, y) , (s, t) , (u, v) respectively, the distance function D has following properties:
 - a. $D(p, q) \geq 0$ $[D(p, q) = 0, \text{ iff } p = q]$
 - b. $D(p, q) = D(q, p)$
 - c. $D(p, z) \leq D(p, q) + D(q, z)$



Distance Measures

The following are the different Distance measures:

a. Euclidean Distance :

$$D_e(p, q) = [(x-s)^2 + (y-t)^2]^{1/2}$$

b. City Block Distance:

$$D_4(p, q) = |x-s| + |y-t|$$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

c. Chess Board Distance:

$$D_8(p, q) = \max(|x-s|, |y-t|)$$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2



Mathematical Operations in DIP

- Array vs. Matrix Operation

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

Array
product
operator

$$A .* B = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Array product

Matrix
product
operator

$$A * B = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Matrix product



Mathematical Operations in DIP

- **Linear vs. Nonlinear Operation**

$$H[f(x, y)] = g(x, y)$$

$$H[a_i f_i(x, y) + a_j f_j(x, y)]$$

$$= H[a_i f_i(x, y)] + H[a_j f_j(x, y)]$$

$$= a_i H[f_i(x, y)] + a_j H[f_j(x, y)]$$

$$= a_i g_i(x, y) + a_j g_j(x, y)$$

Additivity

Homogeneity

H is said to be a **linear operator**;

H is said to be a **nonlinear operator** if it does not meet the above qualification.



Arithmetic Operations

- Arithmetic operations between images are array operations. The four arithmetic operations are denoted as

$$s(x,y) = f(x,y) + g(x,y)$$

$$d(x,y) = f(x,y) - g(x,y)$$

$$p(x,y) = f(x,y) \times g(x,y)$$

$$v(x,y) = f(x,y) \div g(x,y)$$



Example: Addition of Noisy Images for Noise Reduction

Noiseless image: $f(x,y)$

Noise: $n(x,y)$ (at every pair of coordinates (x,y) , the noise is uncorrelated and has zero average value)

Corrupted image: $g(x,y)$

$$g(x,y) = f(x,y) + n(x,y)$$

Reducing the noise by adding a set of noisy images, $\{g_i(x,y)\}$

$$\bar{g}(x,y) = \frac{1}{K} \sum_{i=1}^K g_i(x,y)$$



Example: Addition of Noisy Images for Noise Reduction

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

$$E \{ \bar{g}(x, y) \} = E \left\{ \frac{1}{K} \sum_{i=1}^K g_i(x, y) \right\}$$

$$= E \left\{ \frac{1}{K} \sum_{i=1}^K [f(x, y) + n_i(x, y)] \right\}$$

$$= f(x, y) + E \left\{ \frac{1}{K} \sum_{i=1}^K n_i(x, y) \right\}$$

$$= f(x, y)$$

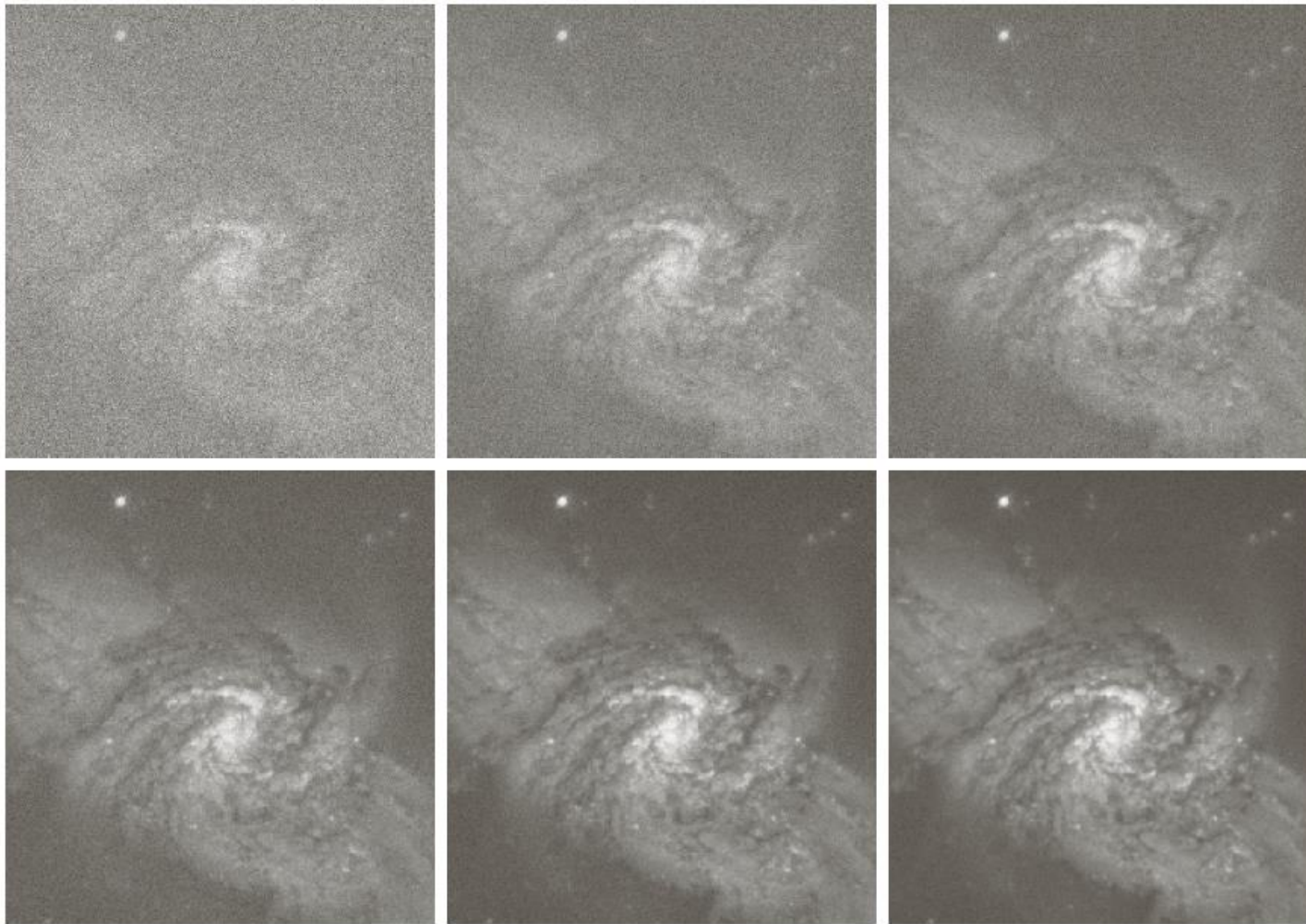
$$\sigma_{\bar{g}(x, y)}^2 = \sigma_{\frac{1}{K} \sum_{i=1}^K g_i(x, y)}^2$$

$$= \sigma_{\frac{1}{K} \sum_{i=1}^K n_i(x, y)}^2 = \frac{1}{K} \sigma_{n(x, y)}^2$$



Example: Addition of Noisy Images for Noise Reduction

- ▶ In astronomy, imaging under very low light levels frequently causes sensor noise to render single images virtually useless for analysis.
- ▶ In astronomical observations, similar sensors for noise reduction by observing the same scene over long periods of time. Image averaging is then used to reduce the noise.



a	b	c
d	e	f

FIGURE 2.26 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)–(f) Results of averaging 5, 10, 20, 50, and 100 noisy images, respectively. (Original image courtesy of NASA.)



An Example of Image Subtraction: Mask Mode Radiography

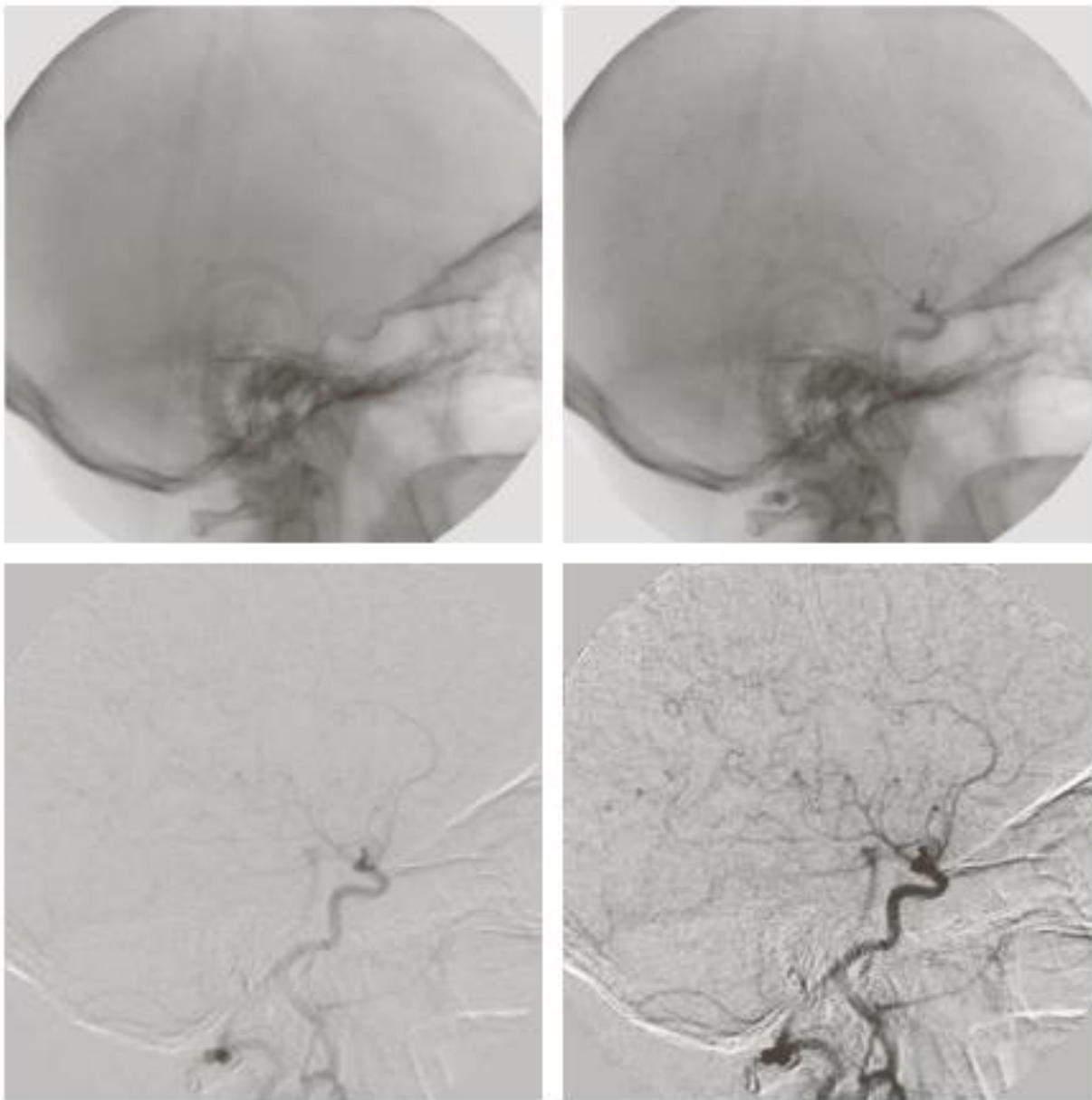
Mask $h(x,y)$: an X-ray image of a region of a patient's body

Live images $f(x,y)$: X-ray images captured at TV rates after injection of the contrast medium

Enhanced detail $g(x,y)$

$$g(x,y) = f(x,y) - h(x,y)$$

The procedure gives a movie showing how the contrast medium propagates through the various arteries in the area being observed.



a	b
c	d

FIGURE 2.28

Digital subtraction angiography.

(a) Mask image.

(b) A live image.

(c) Difference

between (a) and

(b). (d) Enhanced difference image.

(Figures (a) and

(b) courtesy of

The Image

Sciences Institute,

University

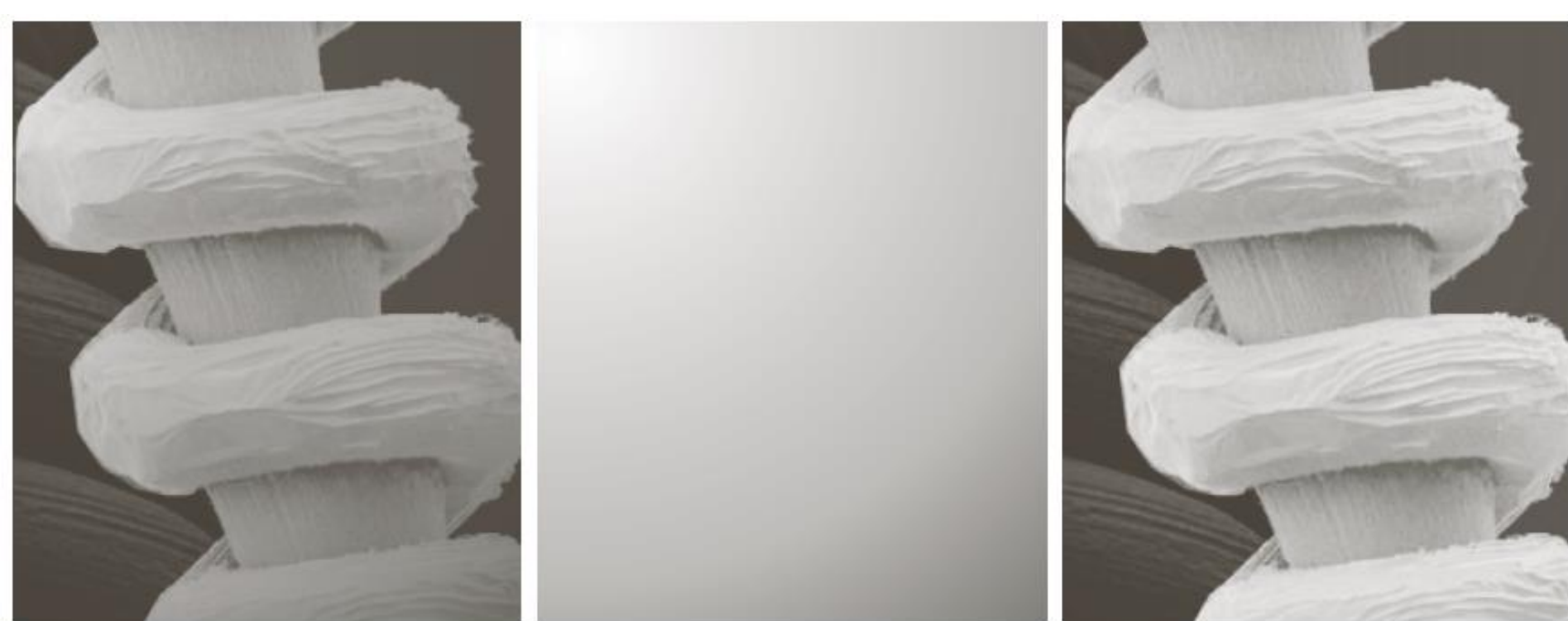
Medical Center,

Utrecht, The

Netherlands.)



An Example of Image Multiplication

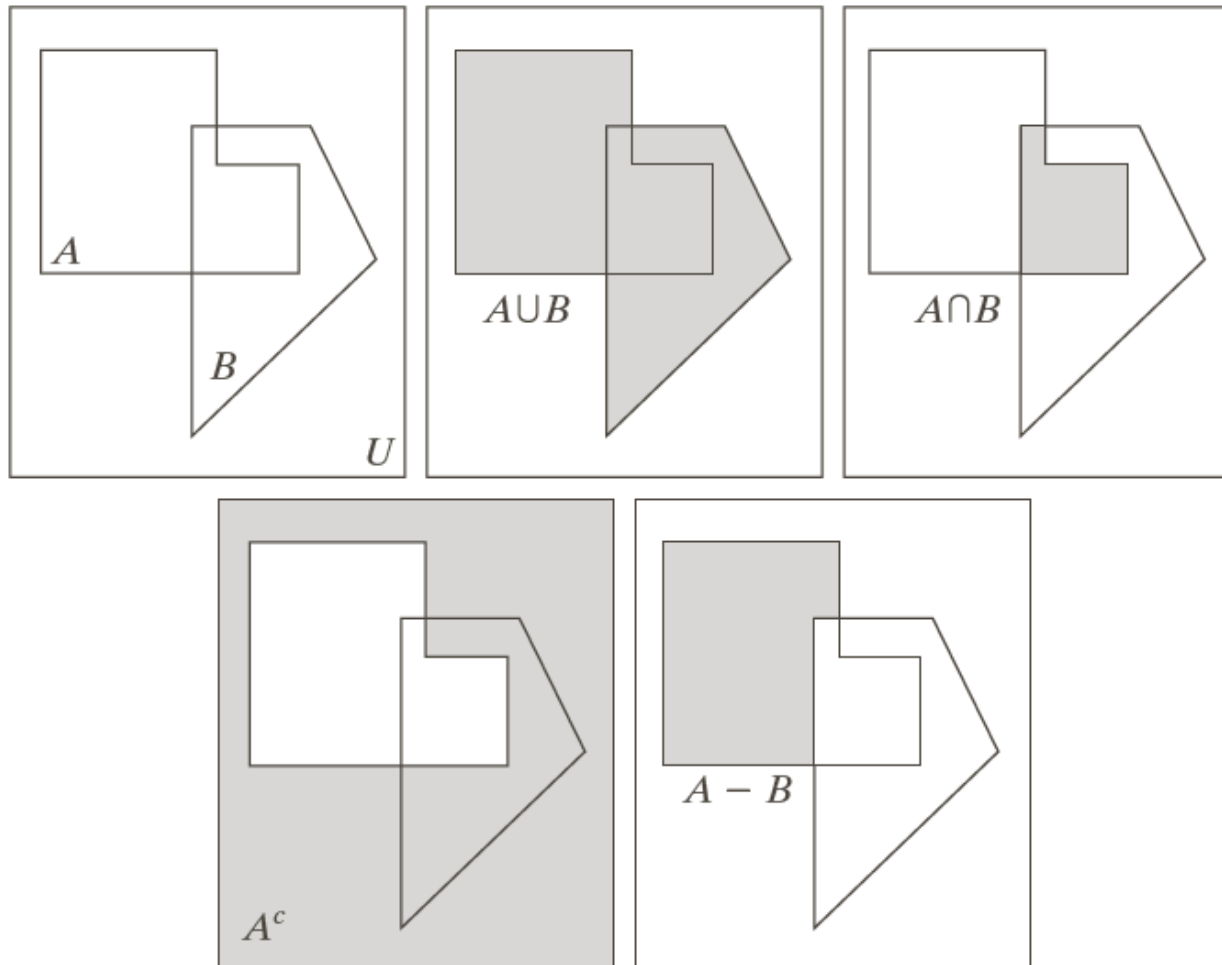


a b c

FIGURE 2.29 Shading correction. (a) Shaded SEM image of a tungsten filament and support, magnified approximately 130 times. (b) The shading pattern. (c) Product of (a) by the reciprocal of (b). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)



Set and Logical Operations



a	b	c
d	e	

FIGURE 2.31

(a) Two sets of coordinates, A and B , in 2-D space. (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B . In (b)–(e) the shaded areas represent the member of the set operation indicated.



Set and Logical Operations

- Let A be the elements of a gray-scale image

The elements of A are triplets of the form (x, y, z) , where x and y are spatial coordinates and z denotes the intensity at the point (x, y) .

$$A = \{(x, y, z) \mid z = f(x, y)\}$$

- The complement of A is denoted A^c

$$A^c = \{(x, y, K - z) \mid (x, y, z) \in A\}$$

$K = 2^k - 1$; k is the number of intensity bits used to represent z



Set and Logical Operations

- The union of two gray-scale images (sets) A and B is defined as the set

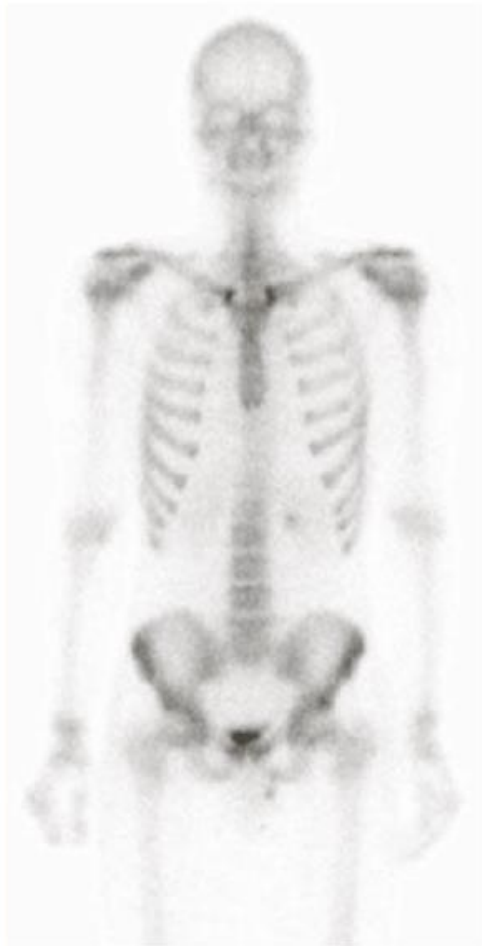
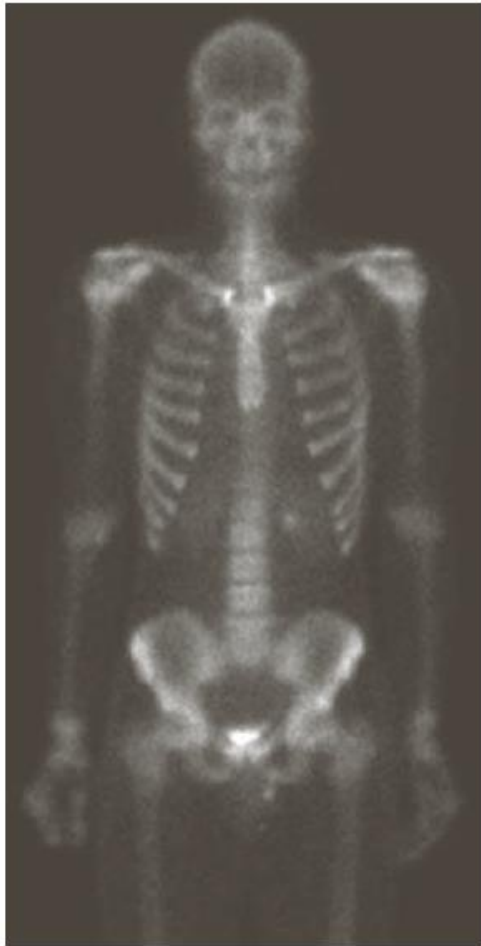
$$A \cup B = \{\max_z(a, b) \mid a \in A, b \in B\}$$



Set and Logical Operations

a b c

FIGURE 2.32 Set operations involving gray-scale images. (a) Original image. (b) Image negative obtained using set complementation. (c) The union of (a) and a constant image. (Original image courtesy of G.E. Medical Systems.)





Set and Logical Operations



FIGURE 2.33

Illustration of logical operations involving foreground (white) pixels. Black represents binary 0s and white binary 1s. The dashed lines are shown for reference only. They are not part of the result.



Spatial Operations

- Single-pixel operations

Alter the values of an image's pixels based on the intensity.

$$s = T(z)$$

e.g.,

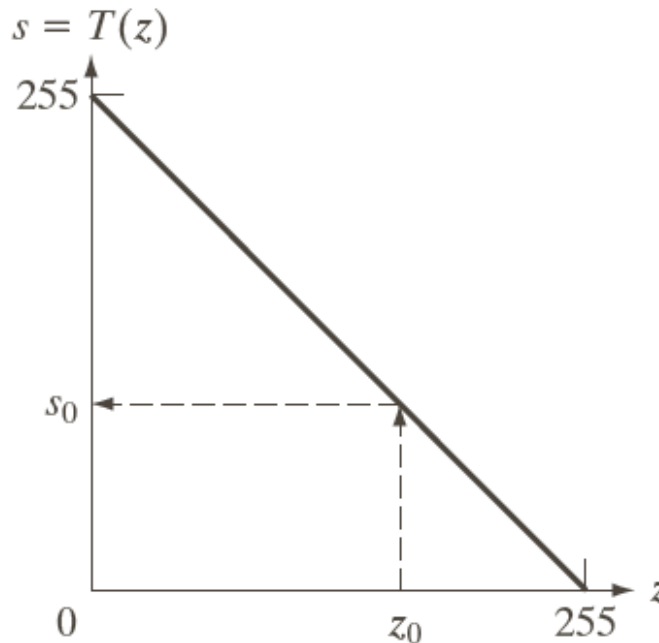
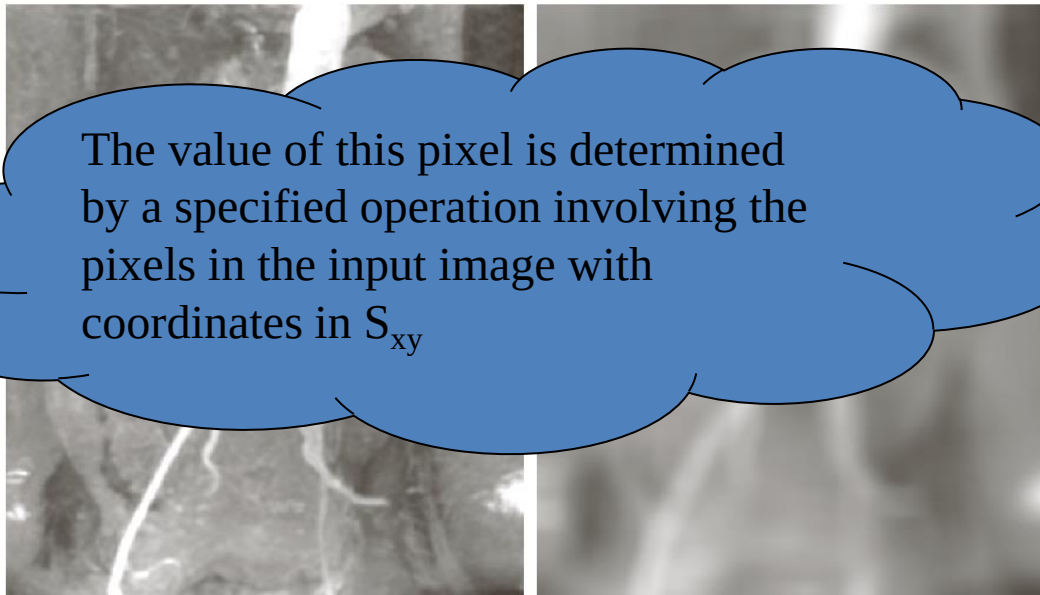
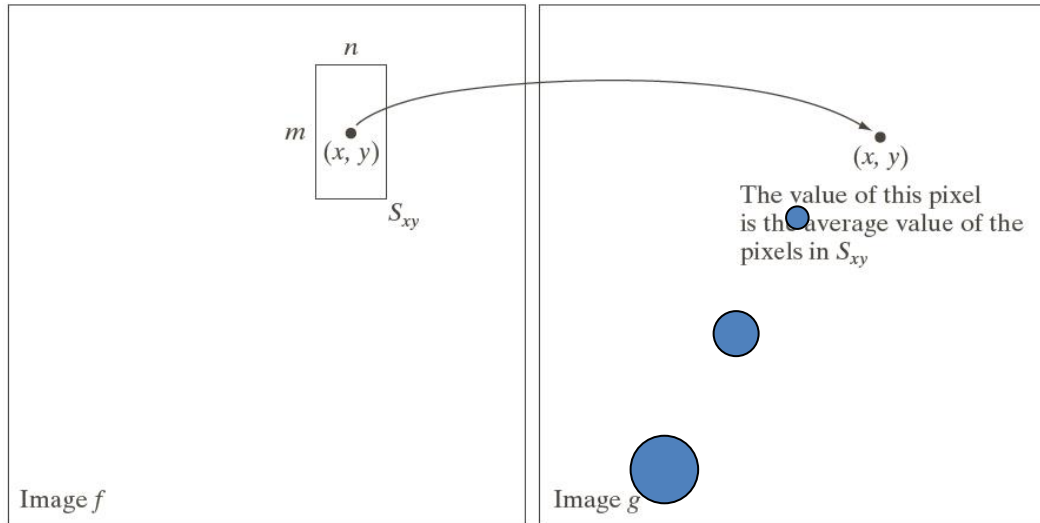


FIGURE 2.34 Intensity transformation function used to obtain the negative of an 8-bit image. The dashed arrows show transformation of an arbitrary input intensity value z_0 into its corresponding output value s_0 .



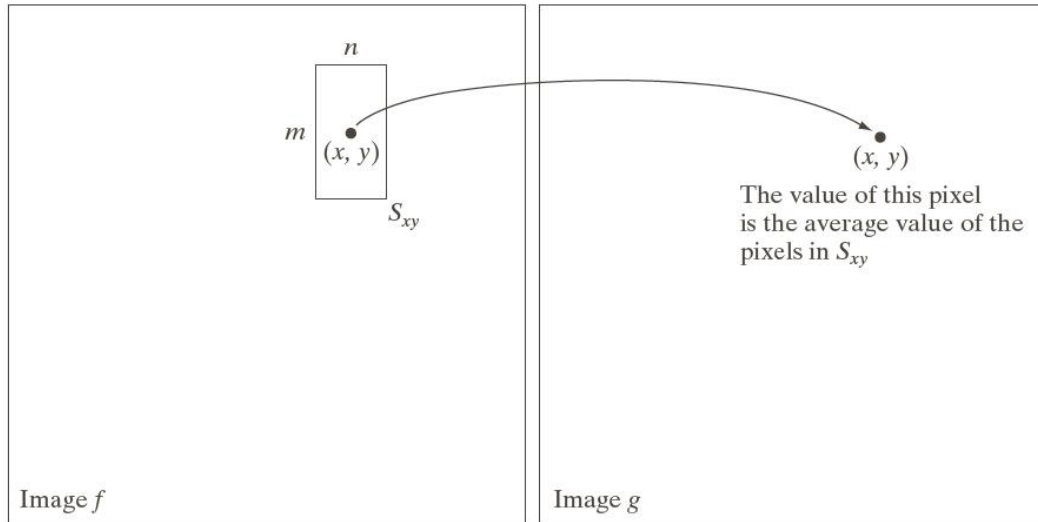
Spatial Operations

- Neighborhood operations





- Neighborhood operations





Geometric Spatial Transformations

- Geometric transformation (rubber-sheet transformation)
 - A spatial transformation of coordinates

$$(x, y) = T\{(v, w)\}$$

— intensity interpolation that assigns intensity values to the spatially transformed pixels.

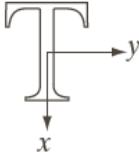
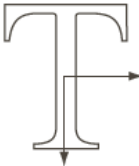
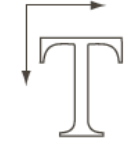
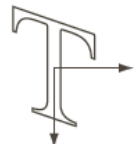
- Affine transform

$$\begin{bmatrix} x & y & 1 \end{bmatrix} = \begin{bmatrix} v & w & 1 \end{bmatrix} \begin{bmatrix} t_{11} & t_{12} & 0 \\ t_{21} & t_{22} & 0 \\ t_{31} & t_{32} & 1 \end{bmatrix}$$



TABLE 2.2

Affine transformations based on Eq. (2.6.–23).

Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \\ y &= w \end{aligned}$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= c_x v \\ y &= c_y w \end{aligned}$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \cos \theta - w \sin \theta \\ y &= v \sin \theta + w \cos \theta \end{aligned}$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$\begin{aligned} x &= v + t_x \\ y &= w + t_y \end{aligned}$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v + s_v w \\ y &= w \end{aligned}$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \\ y &= s_h v + w \end{aligned}$	



Intensity Assignment

- Forward Mapping

$$(x, y) = T\{(v, w)\}$$

It's possible that two or more pixels can be transformed to the same location in the output image.

- Inverse Mapping

$$(v, w) = T^{-1}\{(x, y)\}$$

The nearest input pixels to determine the intensity of the output pixel value. Inverse mappings are more efficient to implement than forward mappings.



Image Rotation and Intensity Interpolation



FIGURE 2.36 (a) A 300 dpi image of the letter T. (b) Image rotated 21° clockwise using nearest neighbor interpolation to assign intensity values to the spatially transformed pixels. (c) Image rotated 21° using bilinear interpolation. (d) Image rotated 21° using bicubic interpolation. The enlarged sections show edge detail for the three interpolation approaches.



Image Registration

- Input and output images are available but the transformation function is unknown.

Goal: estimate the transformation function and use it to register the two images.

- One of the principal approaches for image registration is to use ***tie points*** (also called ***control points***)
 - The corresponding points are known precisely in the input and output (**reference**) images.



Image Registration

- A simple model based on bilinear approximation:

$$\begin{cases} x = c_1 v + c_2 w + c_3 vw + c_4 \\ y = c_5 v + c_6 w + c_7 vw + c_8 \end{cases}$$

Where (v, w) and (x, y) are the coordinates of tie points in the input and reference images.

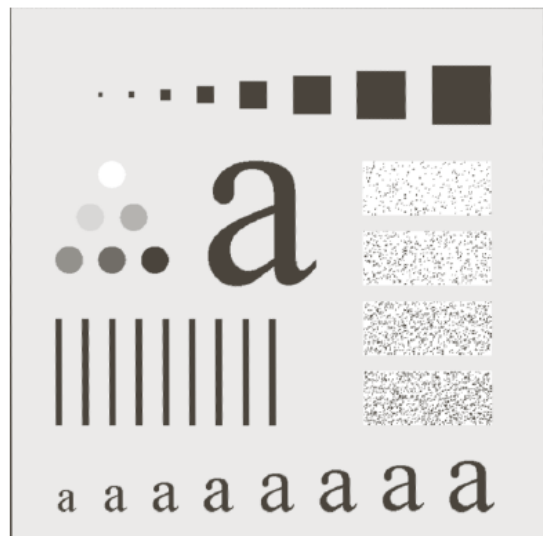


FIGURE 2.37
Image registration.
(a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners.
(c) Registered image (note the errors in the borders).
(d) Difference between (a) and (c), showing more registration errors.



Image Transform

- A particularly important class of 2-D linear transforms, denoted $T(u, v)$

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) r(x, y, u, v)$$

where $f(x, y)$ is the input image,

$r(x, y, u, v)$ is the *forward transformation kernel*,

variables u and v are the transform variables,

$u = 0, 1, 2, \dots, M-1$ and $v = 0, 1, \dots, N-1$.



Image Transform

- Given $T(u, v)$, the original image $f(x, y)$ can be recovered using the inverse transformation of $T(u, v)$.

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) s(x, y, u, v)$$

where $s(x, y, u, v)$ is the *inverse transformation kernel*,
 $x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, \dots, N-1$.



Image Transform

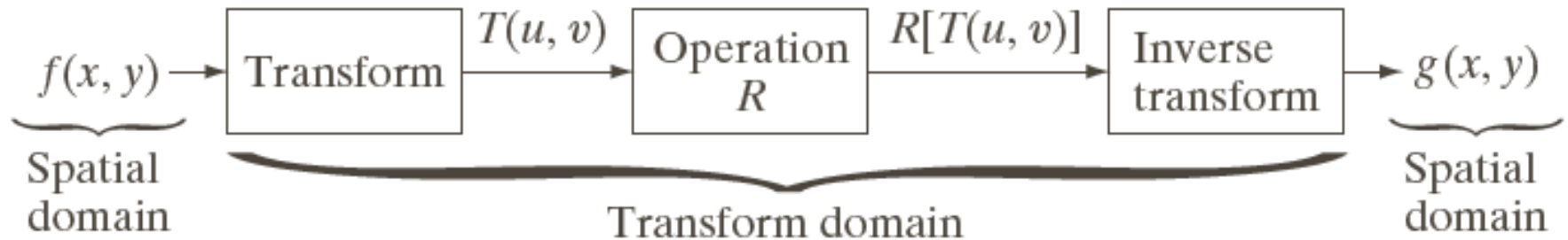
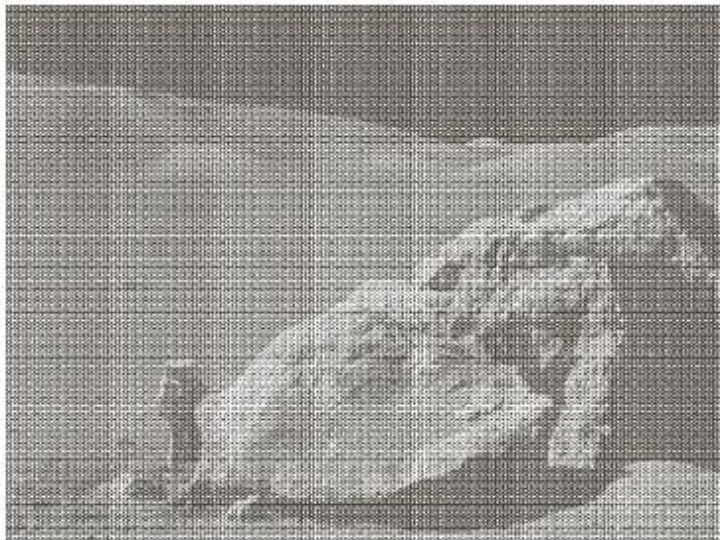


FIGURE 2.39

General approach
for operating in
the linear
transform
domain.



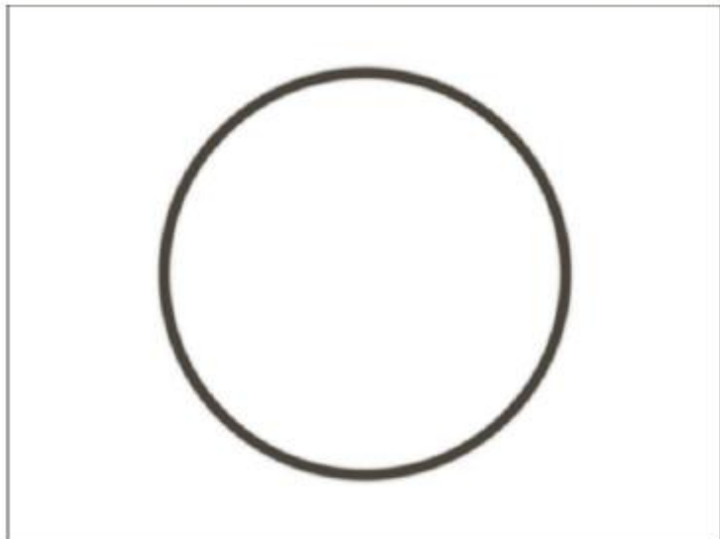
Image Denoising by Using DCT Transform



a	b
c	d

FIGURE 2.40

(a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy responsible for the interference. (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)





Forward Transform Kernel

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) r(x, y, u, v)$$

The kernel $r(x, y, u, v)$ is said to be SEPERABLE if

$$r(x, y, u, v) = r_1(x, u) r_2(y, v)$$

In addition, the kernel is said to be SYMMETRIC if $r_1(x, u)$ is functionally equal to $r_2(y, v)$, so that

$$r(x, y, u, v) = r_1(x, u) r_1(y, u)$$



The Kernels for 2-D Fourier Transform

The *forward* kernel

$$r(x, y, u, v) = e^{-j2\pi(ux/M + vy/N)}$$

Where $j = \sqrt{-1}$

The *inverse* kernel

$$s(x, y, u, v) = \frac{1}{MN} e^{j2\pi(ux/M + vy/N)}$$



2-D Fourier Transform

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) e^{j2\pi(ux/M + vy/N)}$$



Probabilistic Methods

Let z_i , $i = 0, 1, 2, \dots, L-1$, denote the values of all possible intensities in an $M \times N$ digital image. The probability, $p(z_k)$, of intensity level z_k occurring in a given image is estimated as

$$p(z_k) = \frac{n_k}{MN},$$

where n_k is the number of times that intensity z_k occurs in the image.

$$\sum_{k=0}^{L-1} p(z_k) = 1$$

The mean (average) intensity is given by

$$m = \sum_{k=0}^{L-1} z_k p(z_k)$$



Probabilistic Methods

The variance of the intensities is given by

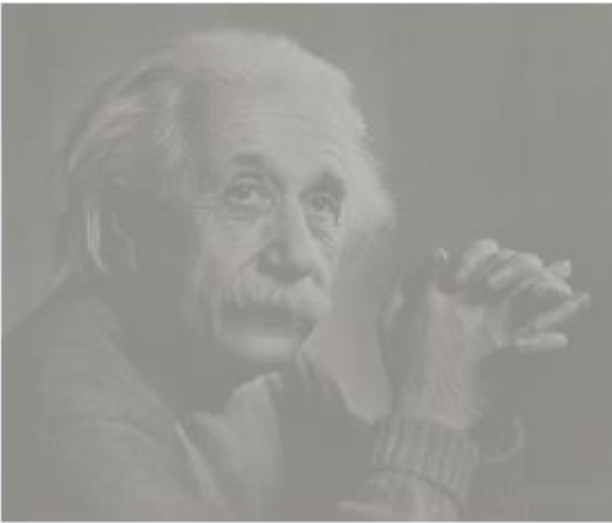
$$\sigma^2 = \sum_{k=0}^{L-1} (z_k - m)^2 p(z_k)$$

The n^{th} moment of the intensity variable z is

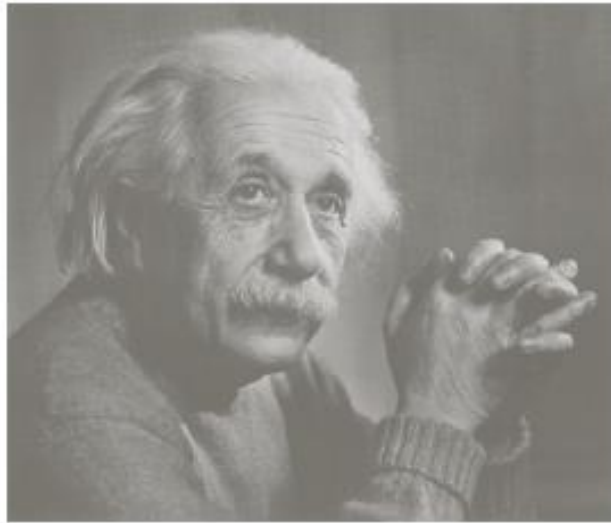
$$u_n(z) = \sum_{k=0}^{L-1} (z_k - m)^n p(z_k)$$



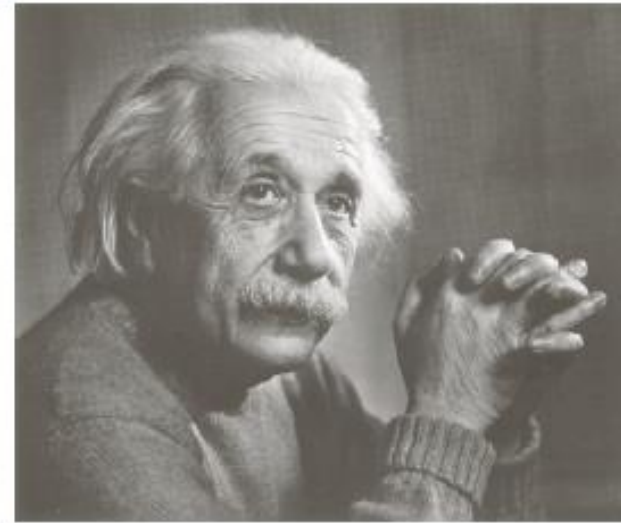
Comparison of Standard Deviation Values



$$\sigma = 14.3$$



$$\sigma = 31.6$$



$$\sigma = 49.2$$



Questions???

Thank You!!!